Web-Based Application For Merchandise Business

By

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Declaration

I hereby attest that this project is entirely my own original effort, as are all related artifacts. It has never been filed for any other degree program before, and it isn't being submitted right now. I certify that all data and information sources utilized in this project have been properly cited. In addition, I have prepared and presented this work by all ethical norms and rules.

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Abstract

Manual business operations often lead to inefficiencies, errors, and customer dissatisfaction. This report outlines the development of a web-based Retail Management System for the CMax Store to address these challenges. The primary goals are to automate administrative tasks, streamline inventory and customer management, and enhance the shopping experience through a modern e-commerce platform.

Key features include real-time inventory tracking, automated order processing, customer feedback collection, and integration with payment gateways. Additional capabilities include personalized product recommendations.

The proposed system introduces a user-friendly interface designed for seamless navigation by both customers and administrators. The platform supports functionalities such as product browsing, Wishlist and cart management, and real-time order tracking. Administrative tools enable efficient inventory control, sales analytics, and customer management.

This report analyzes the current system, details the proposed solution's features and design, and discusses the expected impact on operational efficiency and customer satisfaction. The adoption of this Retail Management System aims to modernize business operations, improve service quality, and foster sustainable growth for the CMax Store.

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1. Introduction

This chapter provides an overview of the business, present business procedures, and current system. The dissertation's business, scope, and bounds are also examined, as well as the drawbacks and problems of the current system and the goals and objectives of the proposed one.

Figure 1 | Official logo of the shop

1.1 Description

CMax Store, located on Kandy Road, Kiribathkumbura, is a prominent merchandising business dedicated to meeting the shopping needs of the local community. The store is owned by Mr Maduranga and is supported by a team of three workers, including one part-time worker and two permanent staff members. As a merchandising business, CMax Store purchases products in bulk or at a lower price and sells them at a higher retail price, thereby generating profit through this markup. The primary purpose of CMax Store is to empower customers with a seamless and personalized shopping experience. The store's vision is to lead Sri Lanka's e-commerce by offering a smart, personalized, and user-friendly platform that enhances the online shopping experience for all.

1.2 Business Process

CMax Store currently operates its essential business processes manually. Below are the current core processes in this merchandising business:

1. **Product Procurement**
2. **Customer Transactions**
3. **Inventory Management**
4. **Customer Feedback Collection**

Core Processes:

1. **Product Procurement Process:**
   The procurement process at CMax Store involves identifying products to purchase based on market demand. The owner places orders with suppliers, manages deliveries, and ensures the availability of products for customers.

2. **Customer Transactions:**
   Customer transactions occur at the physical store, where customers browse products, make selections, and pay for their purchases. The cashier processes payments and issues receipts.

3. **Inventory Management:**
   Inventory is managed manually, with staff keeping track of stock levels and product sales to ensure optimal inventory control and reordering.

4. **Customer Feedback Collection:**
   The store collects customer feedback manually through suggestion boxes or direct conversations, which helps to improve service quality.

1.3 Problems and Weaknesses

- Reliance on Facebook, which is not optimized for efficient e-commerce.
- Inefficient product discovery, requiring extensive scrolling.
- Lack of an integrated online ordering system, requiring manual customer-vendor communication via WhatsApp.
- Manual order confirmations and information gathering.
- Limited communication options for order updates.
- Manual stock tracking, prone to errors.
- Time-consuming process of managing individual product posts.
- Inadequate customer support lacking professionalism and integration typical of dedicated e-commerce systems.

1.4 Aims and Objectives of the project

Aim:
This project aims to enhance the operational efficiency and customer experience at the CMax Store by developing a comprehensive and user-friendly web-based Retail Management System. This system will automate various administrative tasks, streamline communication with customers, and provide valuable insights for informed decision-making.

Objectives:

- **Automation of Administrative Processes:**
  Develop features within the Retail Management System to automate customer transactions, inventory tracking, and staff management.
- **Efficient Customer Interaction:**
  Implement a digital platform for collecting customer feedback and suggestions to improve service quality.
- **Streamlined Inventory Management:**
  Enable real-time tracking of stock levels and automate reordering processes to maintain optimal inventory.
- **User-Friendly Interface:**
  Focus on creating an intuitive user experience for all stakeholders, including customers and staff.

1.5 Scope of the project

The CMax online store will offer functionalities such as product browsing, searching, shopping cart and Wishlist management, order placement and tracking, and notifications. It will include an admin panel for managing products, inventory, users, and orders. To differentiate itself from typical e-commerce sites, the platform will feature an AI-powered chatbot, simplified AI-based product recommendations, gamified loyalty programs, and customizable productdisplays.

2. Current System Analysis

CMax's current process involves customers browsing products on Facebook, which lacks efficient search and filtering options. Orders are placed through WhatsApp, requiring customers to manually share product and shipping details with vendors. Notifications, including order confirmations and tracking information, are also sent manually through WhatsApp, creating delays and room for error. Vendors manage inventory by manually updating Facebook posts, making stock tracking inefficient and error prone. Finally, orders are dispatched with Koombiyo Delivery without integration, further complicating order handling and tracking.

2.1 Use Case Diagram for the Existing Process

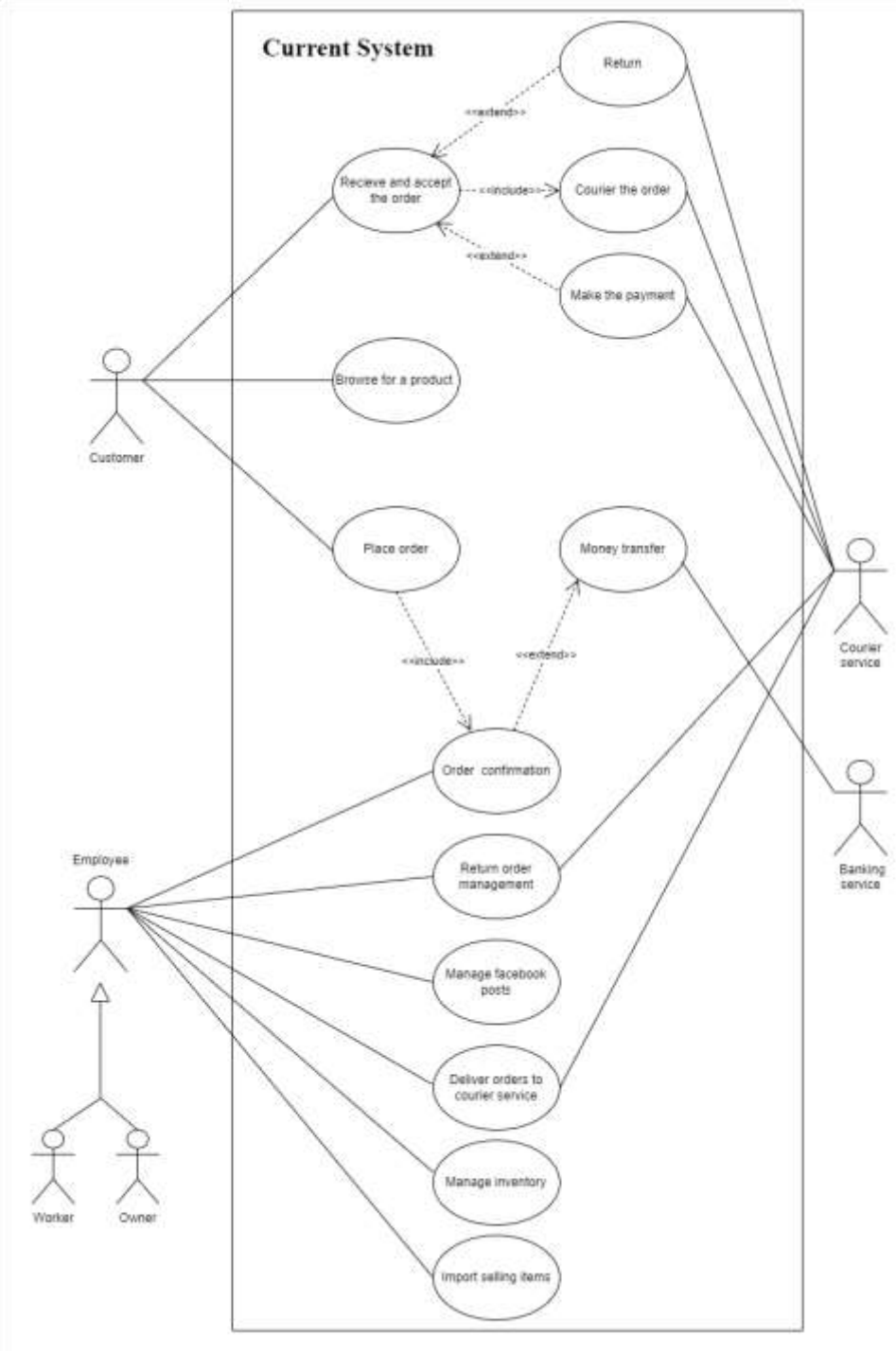

Figure 2 | User Case Diagram for The Existing System

3. Proposed System

3.1 Overall Description

The proposed system is a comprehensive web-based Retail Management System designed to streamline operations at CMax Store. This system offers an integrated platform that automates various administrative processes, enhances customer interactions, and improves inventory management. By transitioning from manual processes to an automated digital solution, the proposed system empowers staff and customers with efficient tools for managing transactions, collecting feedback, and overseeing stock levels.

3.2 Proposed System Functions Overview

The product is designed with several key functions that cater to the operational needs of the CMax Store:

- **Customer Transaction Management:** This allows staff to process customer purchases quickly and efficiently, generate digital receipts, and update inventory automatically.
- **Inventory Management:** Provides real-time tracking of stock levels, alerts for low inventory, and automated reordering processes to ensure product availability.
- **Customer Feedback Collection:** Facilitates the gathering of customer feedback through digital forms and suggestion boxes, enabling analysis and implementation of improvements based on customer insights.
- **Product Management:** Streamlines inventory level tracking, and inventory management, ensuring efficient product management.
- **Reporting and Analytics:** Generate reports on sales, inventory turnover, and customer feedback, providing insights for informed decision-making and strategy development.

3.3 Design and Implementation Constraint

While developing the product, several design and implementation constraints must be considered:

- **Budget Constraints:**
  The project must remain within the allocated budget for software development, which may limit the choice of technologies or additional features.
- **Time Constraints:**
  Given the need to implement the system promptly to enhance business operations, the development timeline may restrict extensive testing or feature expansion.
- **Technical Skills:**
  The technical proficiency of the staff may affect the adoption and utilization of the new system. Training sessions will be necessary to ensure staff are comfortable using the product.
- **Internet Connectivity:**
  As a cloud-based solution, consistent and reliable internet access is essential for the system's functionality. Any disruptions may hinder operations.
- **Data Security:**
  Ensuring the security and privacy of customer data is paramount. Compliance with relevant data protection regulations must be prioritized during development.

3.4 Business System Options Comparison

Business System Options (BSOs) for the CMax Store are alternative approaches or solutions that could meet the business requirements and solve the identified problems. These options typically vary in terms of technology, complexity, cost, and integration levels.

Identified BSOs,

1. *Custom Web Application* - Build a fully customized web application tailored to CMax Store's specific needs, including inventory management, customer interaction, and order processing. Include additional features such as AI-driven product recommendations, loyalty programs, and real-time analytics.
2. *Off-the-Shelf E-commerce Solutions* -Use pre-built e-commerce platforms like Shopify, WooCommerce, or BigCommerce. Customize these platforms to incorporate CMax Store's specific branding and functionality needs.
3. *Cloud-Based SaaS Solutions* - Leverage SaaS platforms like Salesforce Commerce Cloud or Zoho Commerce. These platforms provide integrated solutions for inventory management, customer interaction, and analytics
4. *Open-Source Platforms* - Use open-source tools such as Magento or OpenCart. These solutions are cost-effective but require technical expertise for implementation and maintenance.

Factor/Requirement	Custom Web App	Off-the-Shelf Solution	Cloud-Based SaaS	Open-Source Platform
Real-time Inventory Tracking	x	x	x	x
Email Verification for Registration	x	x	x	x
Product Browsing and Searching	x	x	x	x
Shopping Cart Functionality	x	x	x	x
Order Tracking and Notifications	x	x	x	x
Integration with Payment Gateways	x	x	x	x
Customer Feedback Collection	x	x	x	x
Analytics and Reporting	x	x	x	x
Scalability to Handle Growth	x		x	
Customization for Business Needs	x			x
Security	x		x	
Maintenance and Support	x		x	x
Technical Expertise	x			x
User Experience	x		x	
Mobile Optimization	x		x	

4. System Features

4.1 User Stories

User stories are a way of capturing and communicating requirements in software development. They are typically written from the perspective of an end-user and provide a simple, high-level description of a feature or functionality that the software should deliver.

User Type	User Story ID	Description
Owner	US-01	As an owner, I want to see reports on sales, so I can understand how the business is doing.
Owner	US-02	As an owner, I want to connect to third-party services such as social media, so I can add more features to the system.
Owner	US-03	As an owner, I want the system to back up data regularly, so I can recover it if needed.
Owner	US-04	As an owner, I want the system to run smoothly, even with many users, so my customers have a good experience.
Owner	US-05	As an owner, I want to track inventory levels, so I can maintain optimal stock for customers.
Owner	US-06	As an owner, I want to manage promotions and discounts, so I can attract more customers.
Owner	US-07	As an owner, I want to generate reports on sales and inventory, so I can make better decisions.
Employee	US-08	As an employee, I want to check stock levels, so I can make sure we have enough products.
Employee	US-09	As an employee, I want to assist customers with their orders, so I can provide excellent service.
Employee	US-10	As an employee, I want to manage customer inquiries efficiently, so I can enhance customer satisfaction.
Existing Customer	US-11	As an existing customer, I want to update my personal information, so my account is always correct.
Existing Customer	US-12	As an existing customer, I want to see my order status, so I know when my items will arrive.
Existing Customer	US-13	As an existing customer, I want to chat with support, so I can get help quickly.

Existing Customer	US-14	As an existing customer, I want to save items in my Wishlist, so I can find them later.
Existing Customer	US-15	As an existing customer, I want to view product reviews, so I can make informed purchasing decisions.
Existing Customer	US-16	As an existing customer, I want to manage my payment methods, so I can choose the most convenient option for my purchases.
Existing Customer	US-17	As an existing customer, I want to request returns or exchanges, so I can resolve issues with my orders.
Existing Customer	US-18	As an existing customer, I want to receive real-time updates about my order status, so I know when to expect my items.
Existing Customer	US-19	As an existing customer, I want a simple checkout process, so I can complete my purchase without problems.
Existing Customer	US-20	As an existing customer, I want to see promotions and discounts, so I can save money on my purchases.
Existing Customer	US-21	As an existing, I want to track my orders, so I can stay updated on my purchases.
New Customer	US-22	As a new customer, I want to create an account easily, so I can start shopping right away.
New Customer	US-23	As a new customer, I want to verify my email after signing up, so I can keep my account secure.
New Customer	US-24	As a new customer, I want to browse products and filter them, so I can find what I want quickly.
New Customer	US-25	As a new customer, I want to browse the product catalog without signing up, so I can see what's available first.
New Customer	US-26	As a new customer, I want to find support resources easily, so I can get help if I have questions.

Table 1 | User Stories

4.2 Requirements

4.2.1 Functional Requirements

Functional Requirements describe the specific functionalities or features that a system must provide. These requirements show what the system should do, and the services it should offer.

Functional requirements mainly focus on,

1. Features and capabilities.
2. Main user interactions.
3. Data handling

Requirement ID	Description	Priority	Type
FR-01	The system shall have email verification for registration.	High	Mandatory
FR-02	The system shall enable users to update personal information and manage saved addresses, payment methods, and communication preferences.	High	Mandatory
FR-03	The system shall have a product catalogue, including product listings, categories, images, prices, and descriptions.	High	Mandatory
FR-04	The system shall have search and filtering functionality (keywords, price, brand, colour, size, etc.).	High	Mandatory
FR-05	The system shall have a shopping cart where users can add, view, update, and remove items.	High	Mandatory
FR-06	The system shall have a checkout process that supports checkout, shipping options, payment processing, and order review.	High	Mandatory
FR-07	The system shall allow users to track orders, view and cancel orders, and request returns or exchanges.	High	Mandatory
FR-08	The system shall have integration with local payment gateways.	High	Mandatory
FR-09	The system shall have inventory management to handle stock levels.	High	Mandatory
FR-10	The system shall have admin capabilities for product, order, inventory, and user management, including reporting capabilities.	High	Mandatory
FR-11	The system shall have real-time order status updates.	Medium	Mandatory
FR-12	The system shall have a Wishlist for saving items for future purchases.	Medium	Mandatory
FR-13	The system shall have product reviews and ratings for users to submit and view.	Medium	Mandatory
FR-14	The system shall have analytics and reporting for sales, user behaviour tracking, and inventory monitoring.	Medium	Mandatory
FR-15	The system should have an AI chatbot for customer support, FAQs, and order tracking.	Medium	Optional
FR-16	The system should integrate with third-party services, such as social media platforms.	Medium	Optional
FR-17	The system should have dynamic product bundling with automated pricing adjustments.	Low	Optional

Table 2 | Functional Requirements

4.2.2 Non-Functional Requirements

Nonfunctional requirements describe the characteristics and qualities that define how a system should operate rather than what it should do. These non-functional requirements ensure the overall success and effectiveness of the system.

Non-functional requirements mainly focus on,

1. Performance requirements
2. Reliability and availability of the system.
3. Scalability maintainability.

Requirement ID	Description	Priority
NFR-01	The system should be able to handle 300 users without performance deterioration.	High
NFR-02	The system should implement industry-standard data encryption for sensitive user information and strong user authentication mechanisms, such as multi-factor authentication (MFA).	High
NFR-03	The system should have an intuitive user interface and be compatible with major browsers.	High
NFR-08	The system should adhere to local data protection laws and copyright regulations for all content.	High
NFR-04	The system should implement error-handling strategies to inform users without exposing sensitive information and process all transactions without data loss or corruption.	Medium
NFR-05	The system should have a minimum uptime of 90% per year, excluding scheduled maintenance, and remain operational despite individual component failures.	Medium
NFR-06	The system should be deployable across different operating systems and cloud platforms without significant changes and function consistently across all major browsers.	Medium
NFR-07	The system should be able to scale horizontally (adding more servers) and vertically (increasing server capacity) and adjust resources based on demand.	Medium
NFR-09	The system should seamlessly integrate with third-party services like payment gateways, shipping providers, and social media platforms.	Medium
NFR-10	The system should have regular backups and a clear recovery plan to restore the system in case of a disaster.	Medium
NFR-11	The system should have promotions and discounts, including coupon codes, loyalty programs, and sales management.	Low

Table 3| Non-Functional Requirements

5. System Design

5.1 Overview of the proposed system: Use case diagram

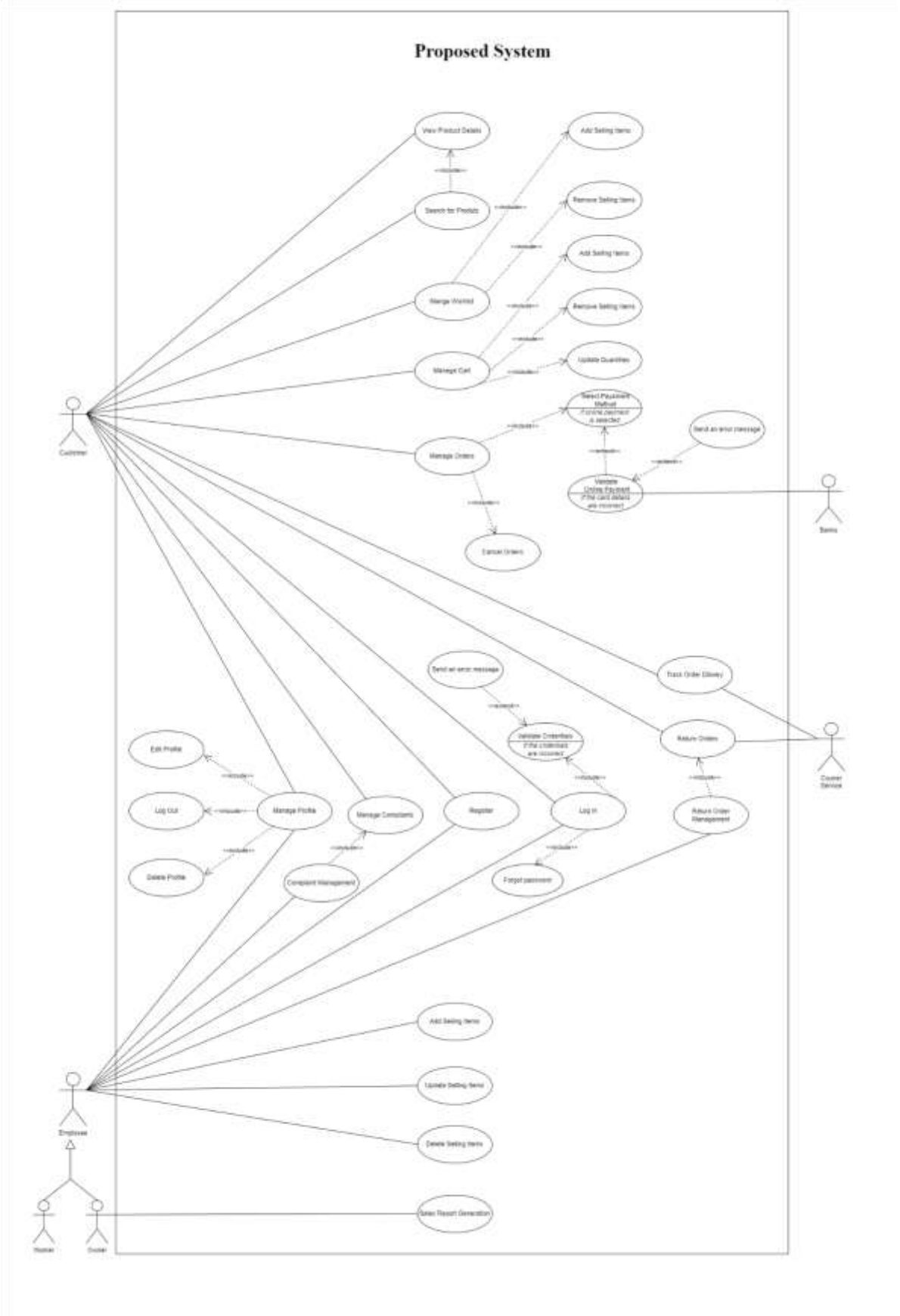

Figure 3 | Use case Diagram for The Proposed System

5.2 Use case descriptions

A use case description is a detailed account that offers an in-depth description of a specific system use case. It demonstrates the actors' interactions with the system. In the particular use case, it describes the precise actions or sequences of events that take place. For developers and stakeholders, use case descriptions offer thorough knowledge. Use case descriptions give a clear explanation of how a specific system component is supposed to operate, which helps to clarify system requirements. By doing this, misconceptions and misinterpretations are prevented.

5.2.1 Register

Name of Use Case	Register
Use Case ID	UCD - 1
Description	The user registers for an account on the CMax Online Store platform to access personalized services.
Actors	User, CMax System
Preconditions	1. The user has internet access. 2. The user has not previously registered an account with the same email.
Postconditions	1. The user receives a confirmation of successful registration. 2. The user can log in with the registered credentials.
Flow	1. The user navigates to the registration page. 2. User enters required details (e.g., name, email, password). 3. The user submits the form. 4. The system verifies details and creates an account. 5. System displays confirmation of registration.
Alternative Flows	If the email is already registered, the system shows an error message.
Exceptions	If there is a system error during registration, the user is notified, and the registration process is halted.
Requirements	1. User email validation is required. 2. Password strength requirements must be met.

Table 4 | Use case description - Register

5.2.2 Login

Name of Use Case	Login
Use Case ID	UCD - 2
Description	Allows users to log in to the CMax Online Store using their registered credentials.
Actors	User, CMax System
Preconditions	The user must have a registered account.
Postconditions	1. The user is logged in and redirected to the homepage. 2. The user has access to personalized account features.
Flow	1. The user navigates to the login page. 2. User enters email and password. 3. The system verifies credentials.

	4. The user gains access if the credentials are correct.
Alternative Flows	If credentials are incorrect, the system displays an error message.
Exceptions	If the system fails during login, an error message is displayed, and the user cannot log in.
Requirements	1. The password should be masked. 2. Failed login attempts should trigger an account lock after a set number of tries.

Table 5 | Use case description - Login

5.2.3 View Product

Name of Use Case	View Product
Use Case ID	UCD - 3
Description	Users can view details of specific products, including description, price, and availability.
Actors	User, CMax System
Preconditions	The user has successfully logged in (if required for view).
Postconditions	Product details are displayed on the screen.
Flow	1. The user selects a product. 2. The system retrieves and displays product details from the database.
Alternative Flows	If the product is out of stock, display a message indicating unavailability.
Exceptions	If the product data fails to load, an error message is displayed, and the user returns to the product list.
Requirements	Product information must be accurate and up to date.

Table 6 | Use case description - View Product

5.2.4 Search

Name of Use Case	Search
Use Case ID	UCD - 4
Description	Users can search for products using keywords to quickly locate items.
Actors	User, CMax System
Postconditions	Search results are displayed based on user input.
Flow	1. User enters search keywords. 2. The system processes the search and retrieves matching results. 3. Results are displayed to the user.
Alternative Flows	If no products match the search, the system displays a message indicating no results found.
Exceptions	If the search fails, an error message is shown, and the user is redirected to the search page.

Requirements	Search should be fast and support multiple filters.

Table 7 | Use case description- Search

5.2.5 Manage Wishlist

Name of Use Case	Manage Wishlist
Use Case ID	UCD - 5
Description	Users can add or remove items from their Wishlist for future reference.
Actors	User, CMax System
Preconditions	The user must be logged in.
Postconditions	The Wishlist is updated according to user selections.
Flow	1. The user selects an item to add to/remove from the Wishlist. 2. The System updates the Wishlist in the user profile.
Exceptions	If the Wishlist update fails, an error message is shown to the user.
Requirements	Wishlist changes should be saved immediately for future access.

Table 8 | Use case description - Manage Wishlist

5.2.6 Manage Cart

Name of Use Case	Manage Cart
Use Case ID	UCD - 6
Description	Users can add, update, or remove items in their shopping cart.
Actors	User, CMax System
Preconditions	The user must be logged in.
Postconditions	The shopping cart is updated based on user actions.
Flow	1. The user selects an item to add/update/remove. 2. System updates cart contents and totals accordingly.
Alternative Flows	If an item is out of stock, show a message indicating unavailability.
Exceptions	If the cart update fails, show an error message and reset the cart state.
Requirements	The system must track real-time inventory levels.

Table 9 | Use case description - Manage Cart

5.2.7 Place Order

Name of Use Case	Place Order
Use Case ID	UCD - 7
Description	Allows users to review their cart, provide details, and place an order.
Actors	User, CMax System, Payment Gateway
Preconditions	The user is logged in and has items in the cart.

Postconditions	Order is placed, payment processed, and confirmation sent to the user.
Flow	1. The user reviews the cart and provides delivery/payment details. 2. System processes payment with Payment Gateway. 3. System sends order confirmation.
Alternative Flows	If payment fails, the user is prompted to retry or use a different method.
Exceptions	If inventory is insufficient, the order cannot be placed, and an error message is shown.
Requirements	Secure payment processing and inventory validation are required.

Table 10 | Use case description- Place Order

5.2.8 Cancel Order

Name of Use Case	Cancel Order
Use Case ID	UCD - 8
Description	Users can cancel an order before it is shipped.
Actors	User, CMax System
Preconditions	The order is in a cancellable state (not yet shipped).
Postconditions	The order is cancelled, and the user is notified.
Flow	1. The user selects an order to cancel. 2. System verifies order status. 3. The system cancels the order and notifies the user.
Alternative Flows	If the order has shipped, the user is informed that cancellation is not possible.
Exceptions	If the cancellation fails, the user is notified of an error.
Requirements	Real-time order status tracking is required.

Table 11 | Use case description - Cancel Order

5.2.9 Order Tracking

Name of Use Case	Order Tracking
Use Case ID	UCD - 9
Description	Allows users to check the status and track the shipment of their orders.
Actors	User, CMax System, Courier Service
Preconditions	The user has an active order with tracking information.
Postconditions	Order status and tracking information are displayed to the user.
Flow	1. The user navigates to the order tracking page. 2. The user enters the order ID. 3. The system retrieves tracking information from the courier. 4. The system displays the current order status and tracking details.

Alternative Flows	If tracking information is not available, the system notifies the user that tracking details are pending.
Exceptions	If the order ID is invalid, an error message is shown, and the user is prompted to re-enter the ID.
Requirements	The system must integrate with the courier service for real-time tracking updates.

Table 12 | Use case description - Order Tracking

5.2.10 Return Order Management

Name of Use Case	Return Order Management
Use Case ID	UCD - 10
Description	Users can initiate a return request for items within an order.
Actors	User, CMax System, Courier Service
Preconditions	1. Order is eligible for return (within the return period).
Postconditions	The return request is processed, and the user receives confirmation.
Flow	1. The user selects the order to return and initiates a return request. 2. The system verifies eligibility and generates a return authorization. 3. The user sends the product to the courier for return processing. 4. System updates return status and notifies the user.
Alternative Flows	If the return period has expired, the user is notified that the return is no longer possible.
Exceptions	If the return process fails at any point, an error message is shown to the user.
Requirements	The system should handle status updates and integrate with the courier service for return processing.

Table 13 | Use case description - Return Order Management

5.2.11 Manage Profile

Name of Use Case	Manage Profile
Use Case ID	UCD - 11
Description	Allows users to manage and update their profile information, including contact details and preferences.
Actors	User, CMax System
Preconditions	The user must be logged in.
Postconditions	Profile information is updated successfully.
Flow	1. User accesses profile settings. 2. User updates details (e.g., name, address, email). 3. The system saves changes and confirms the update.

Exceptions	If the profile update fails, an error message is shown, and the user is prompted to retry.
Requirements	The system should ensure that profile information is stored securely.

Table 14 | Use case description - Manage Profile

5.2.12 Add Complaint

Name of Use Case	Add Complaint
Use Case ID	UCD - 12
Description	Allows users to submit complaints regarding orders or products directly through the system.
Actors	User, CMax System
Preconditions	The user is logged in and has an active order.
Postconditions	The complaint is logged in the system, and the user receives a confirmation message.
Flow	1. The user navigates to the complaints section. 2. The user fills out the complaint form. 3. The system logs the complaint and confirms it to the user.
Exceptions	If complaint submission fails, an error message is shown, and the user is prompted to retry.
Requirements	The system should ensure complaints are routed to the customer service team.

Table 15 | Use case description - Add Complaint

5.2.13 Manage Complaint

Name of Use Case	Manage Complaint
Use Case ID	UCD - 13
Description	Customer support can review and respond to complaints submitted by users.
Actors	Customer Support, CMax System
Preconditions	A user has submitted a complaint.
Postconditions	The complaint is reviewed and addressed, and the user is notified of the resolution.
Flow	1. Support staff accesses the complaints dashboard. 2. Staff reviews complaint details. 3. Staff responds to the complaint or escalates. 4. The system notifies the user of the response.
Exceptions	If the complaint cannot be resolved, the support team will escalate the issue, and the user will be notified of the delay.
Requirements	The system should allow complaint status tracking and staff responses.

Table 16 | Use case description - Manage Complaint

5.2.14 Add Item

Name of Use Case	Add Item
Use Case ID	UCD - 14
Description	Admins or vendors can add new products to the inventory.
Actors	Vendor/Admin, CMax System
Preconditions	The vendor is logged in with the required permissions.
Postconditions	The new item is successfully added to the product catalogue.
Flow	1. Vendor selects "Add Item" in the product management section. 2. Vendor enters item details. 3. The system saves the product in the catalogue.
Alternative Flows	If item details are incomplete, the system shows an error message.
Exceptions	If there's a system error, the item cannot be added, and the vendor is notified.
Requirements	Item details must meet data validation rules before saving.

Table 17 | Use case description - Add Item

5.2.15 Update Item

Name of Use Case	Update Item
Use Case ID	UCD - 15
Description	Vendors can update the details of existing products in the catalogue.
Actors	Vendor/Admin, CMax System
Preconditions	The vendor is logged in with necessary permissions and the product exists in the system.
Postconditions	Product details are updated in the catalogue.
Flow	1. Vendor selects an item to update. 2. Vendor edits product details. 3. The system saves the changes to the catalogue.
Alternative Flows	If the product is not found, show an error message.
Exceptions	If the update fails, an error message is shown, and the vendor is prompted to retry.
Requirements	The system should enforce data validation rules for updated information.

Table 18 | Use case description - Update Item

5.2.16 Delete Item

Name of Use Case	Delete Item
Use Case ID	UCD - 16
Description	Vendors can delete an item from the product catalogue if it's no longer available or required.
Actors	Vendor/Admin, CMax System
Preconditions	The vendor is logged in with delete permissions, and the item exists in the catalogue.
Postconditions	The item is removed from the catalogue, and users can no longer view it.
Flow	1. Vendor selects an item to delete. 2. The vendor confirms the deletion. 3. The system removes the item from the catalogue.
Alternative Flows	If the item is linked to active orders, display a message that it cannot be deleted.
Exceptions	If the deletion fails, an error message is shown to the vendor.
Requirements	The system should ensure no active dependencies exist on the item before allowing deletion.

Table 19 | Use case description - Delete Item

5.3 Activity Diagrams

5.3.1 Register Activity

The user accesses the registration page, fills out the necessary details (e.g., name, email, password), and submits the form. The system checks the information for completeness and uniqueness, creating the account if valid, and confirming registration to the user. If the email is already registered, the system notifies the user to try a different email.

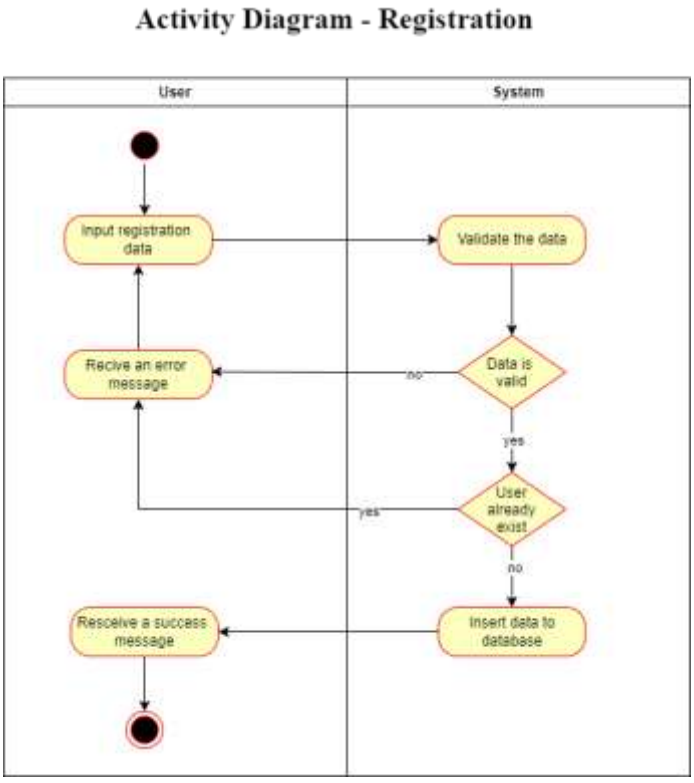

Figure 4 | Activity Diagram - Registration

5.3.2 Login Activity

The user enters their login credentials on the login page and submits. The system validates the credentials; if correct, the user gains access and is redirected to the homepage. If credentials are incorrect, an error message is shown, allowing the user to retry.

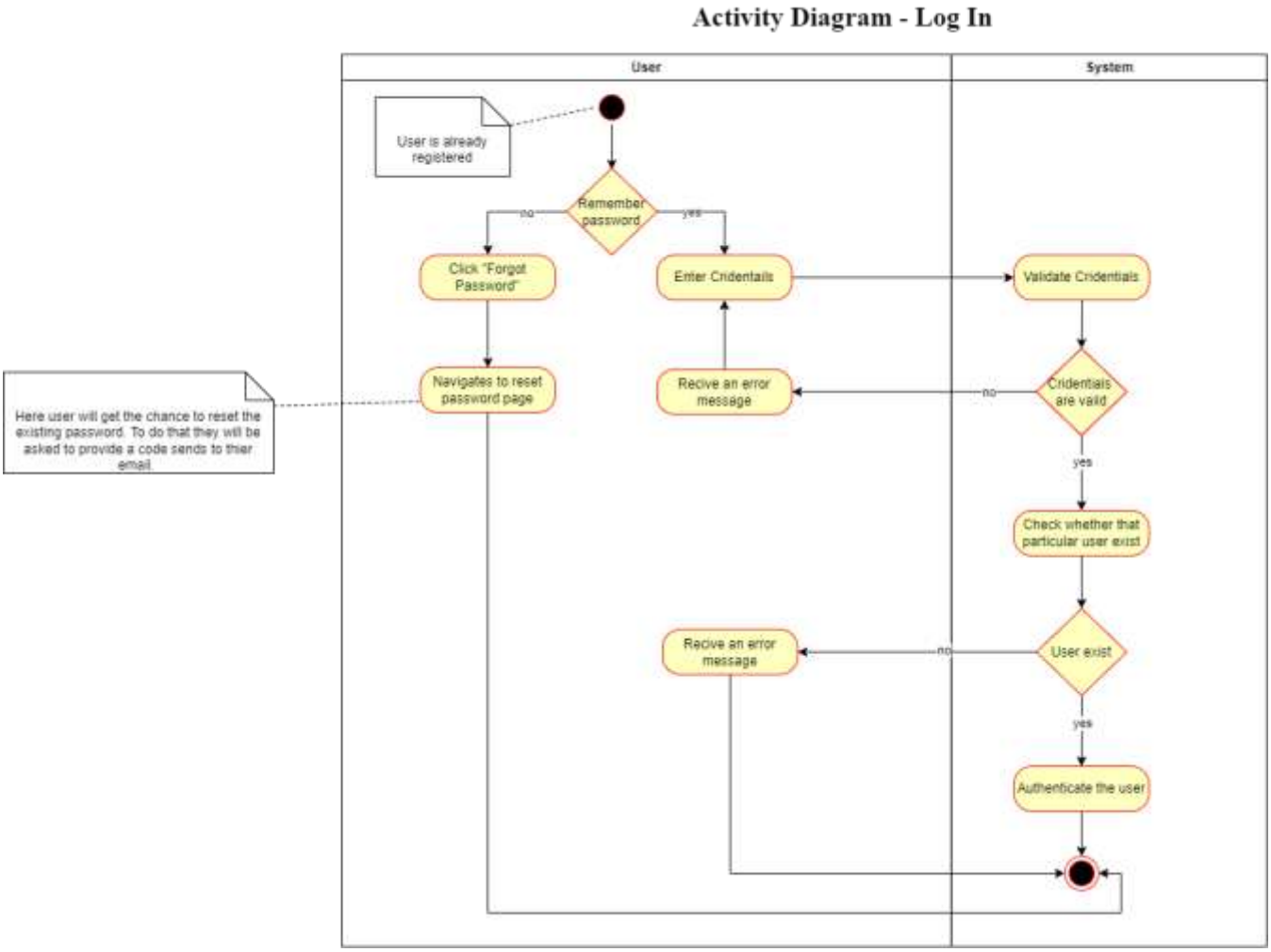

Figure 5 | Activity Diagram – Log In

5.3.3 View Product Activity

When a user selects a product, the system retrieves and displays detailed information (description, price, availability). If the product is out of stock, the system notifies the user. In case of data retrieval errors, an error message is shown, and the user is returned to the product list.

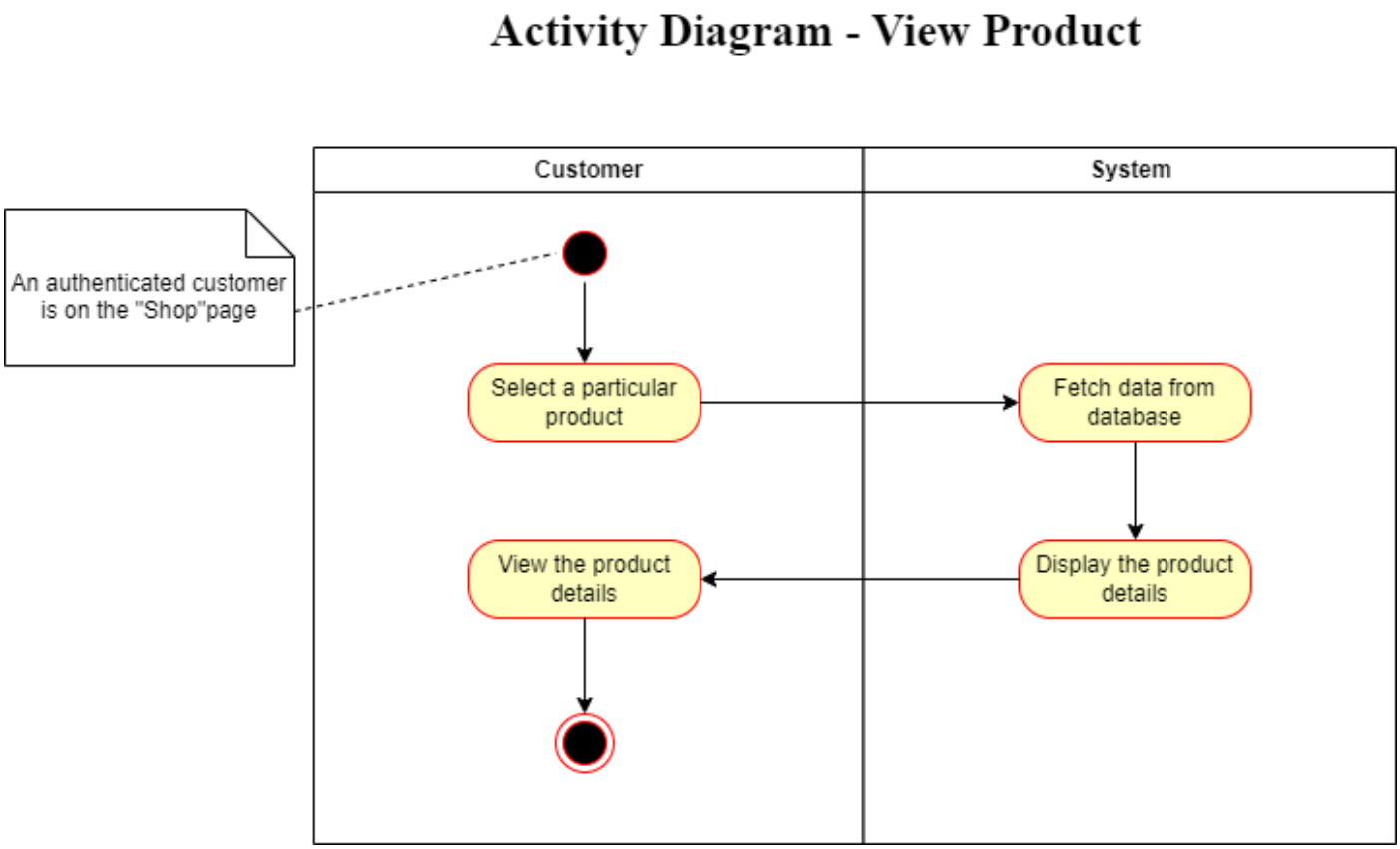

Figure 6 | Activity Diagram - View Product

5.3.4 Search Activity

The user enters search keywords and submits the query. The system retrieves products matching the keywords and displays them. If no matches are found, the system shows a “No Results” message, or an error message if there’s a system issue, prompting the user to retry.

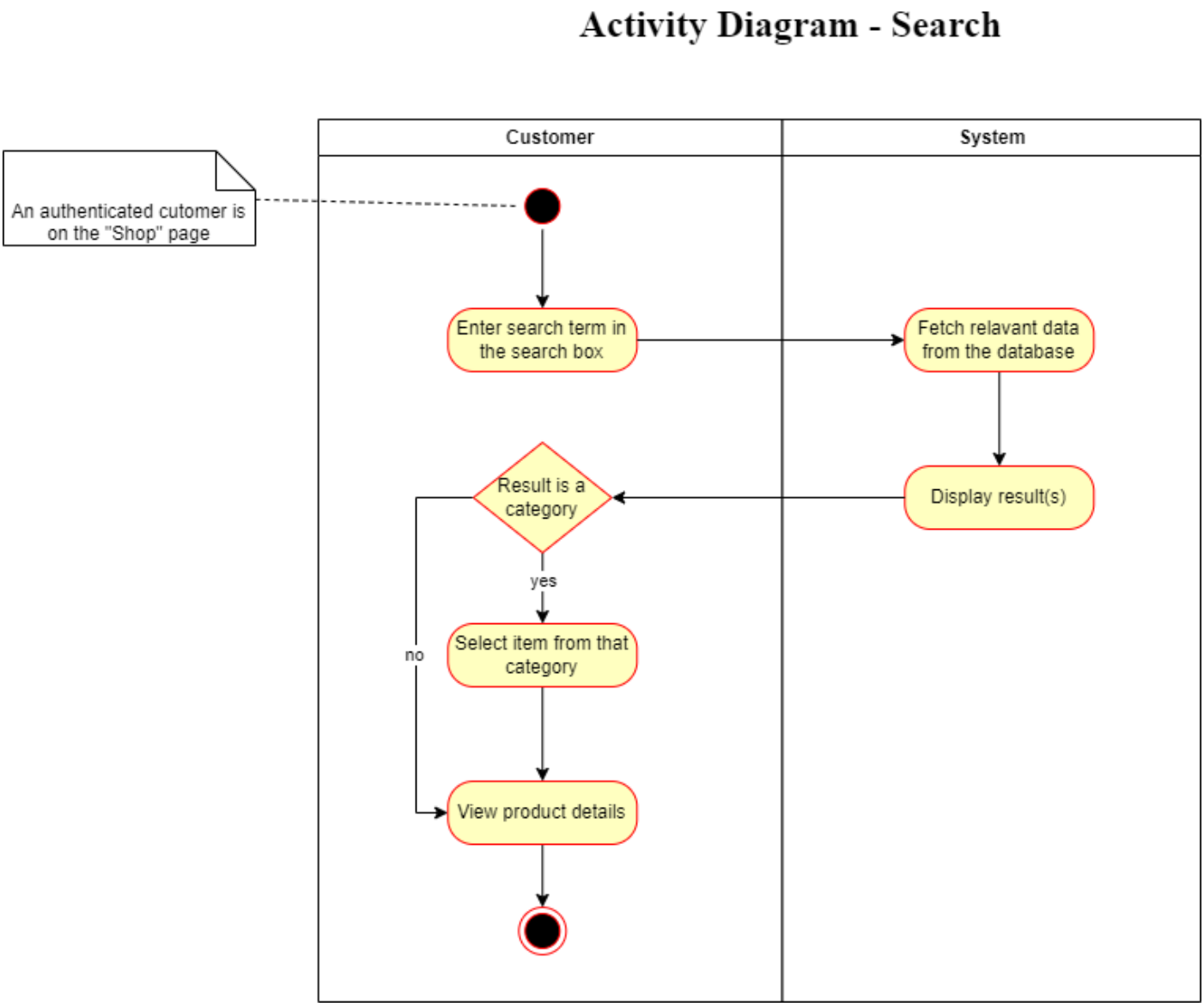

Figure 7 | Activity Diagram - Search

5.3.5 Manage Wishlist Activity

The user adds or removes items from their Wishlist by selecting specific products. The system updates the user's Wishlist and saves changes. If adding an item fails, the system notifies the user and may suggest removing items if the Wishlist has reached its limit.

Activity Diagram - Manage Wishlist

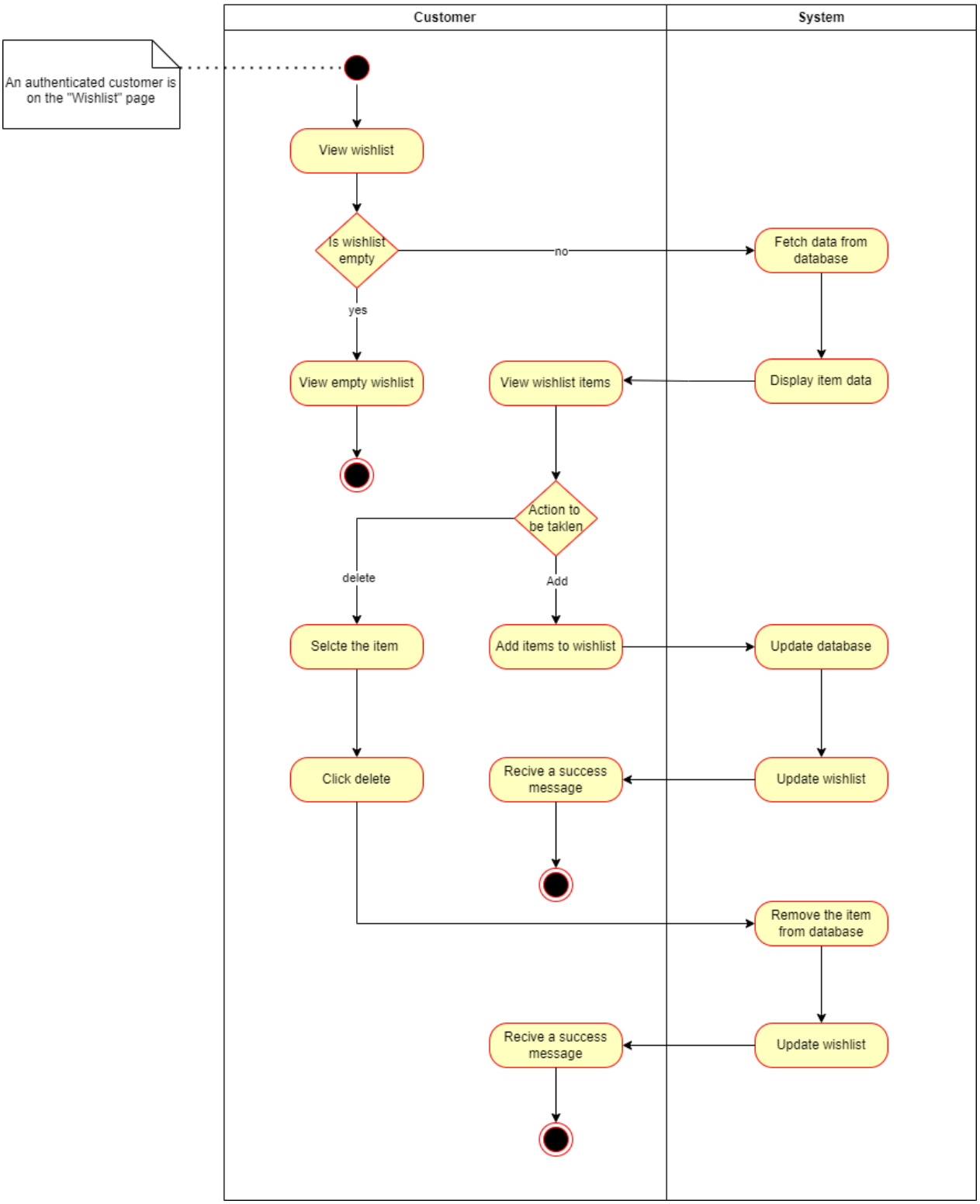

Figure 8 | Activity Diagram - Manage Wishlist

5.3.6 Manage Cart Activity

The user adds, updates, or removes items in the cart, with the system adjusting the cart contents and recalculating the total cost. If an item is out of stock, the system alerts the user, or an error message appears if a cart update fails, restoring the previous cart state.

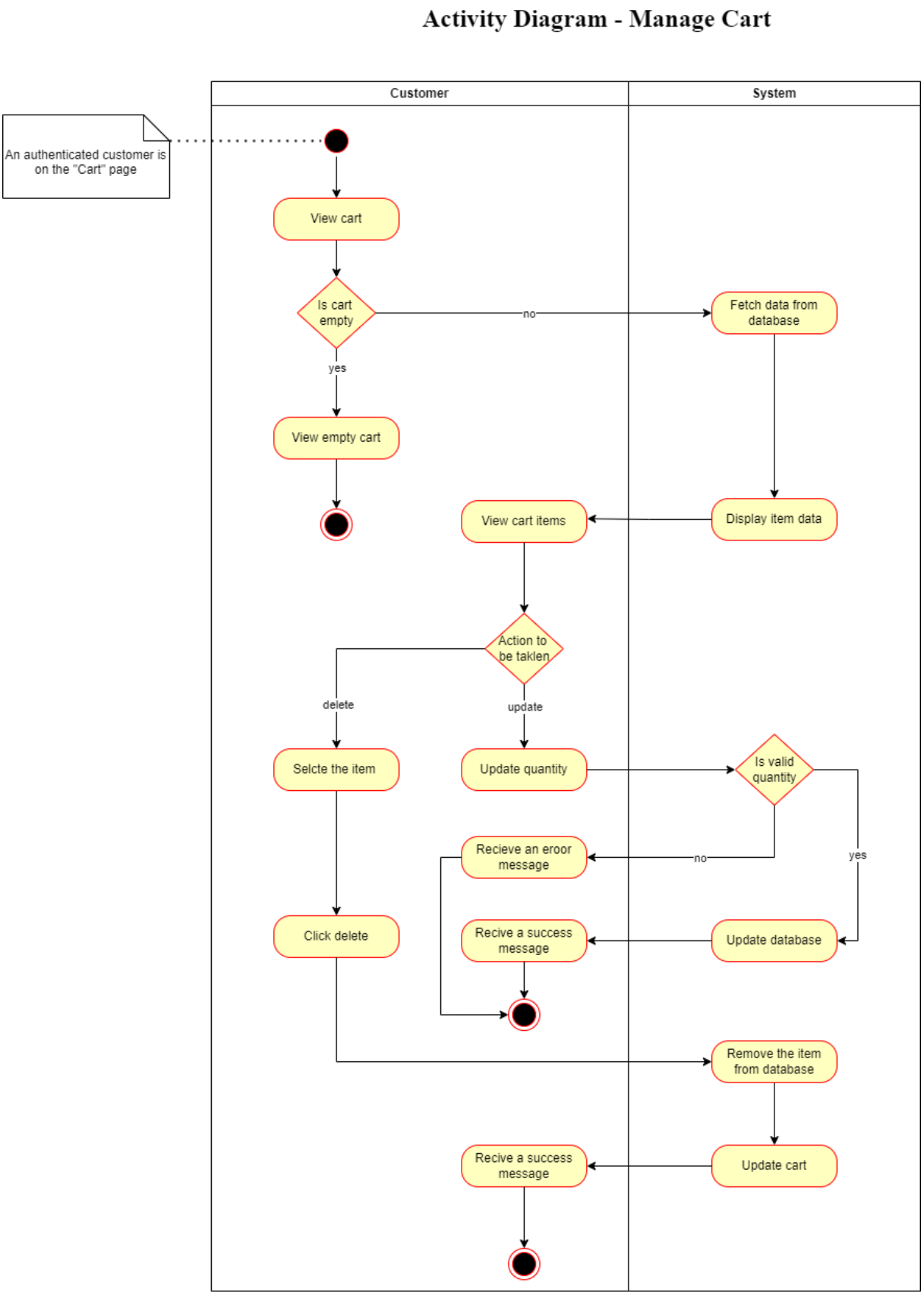

Figure 9 | Activity Diagram - Manage Cart

5.3.7 Place Order Activity

The user reviews the cart, enters payment and shipping details, and submits the order. The system validates stock availability, processes payment via a payment gateway, and confirms the order. If payment fails, the user is prompted to retry, or if an item is unavailable, the user is asked to adjust the cart.

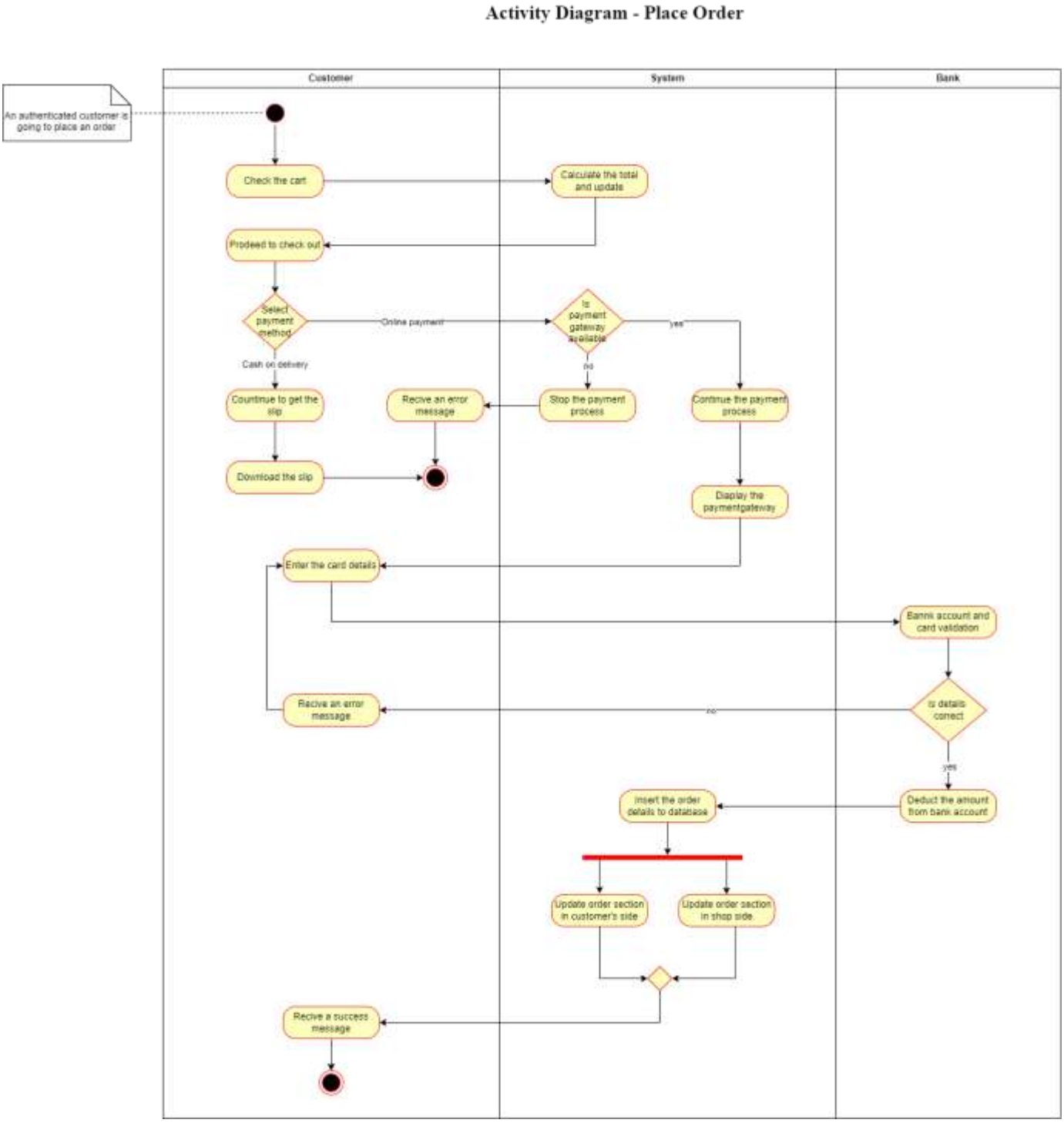

Figure 10 | Activity Diagram - Place Order

5.3.8 Cancel Order Activity

The user selects an active order to cancel. The system verifies eligibility (e.g., the order hasn't shipped), cancels it if valid, and confirms the cancellation to the user. If the order has shipped, cancellation is denied, and the user is notified.

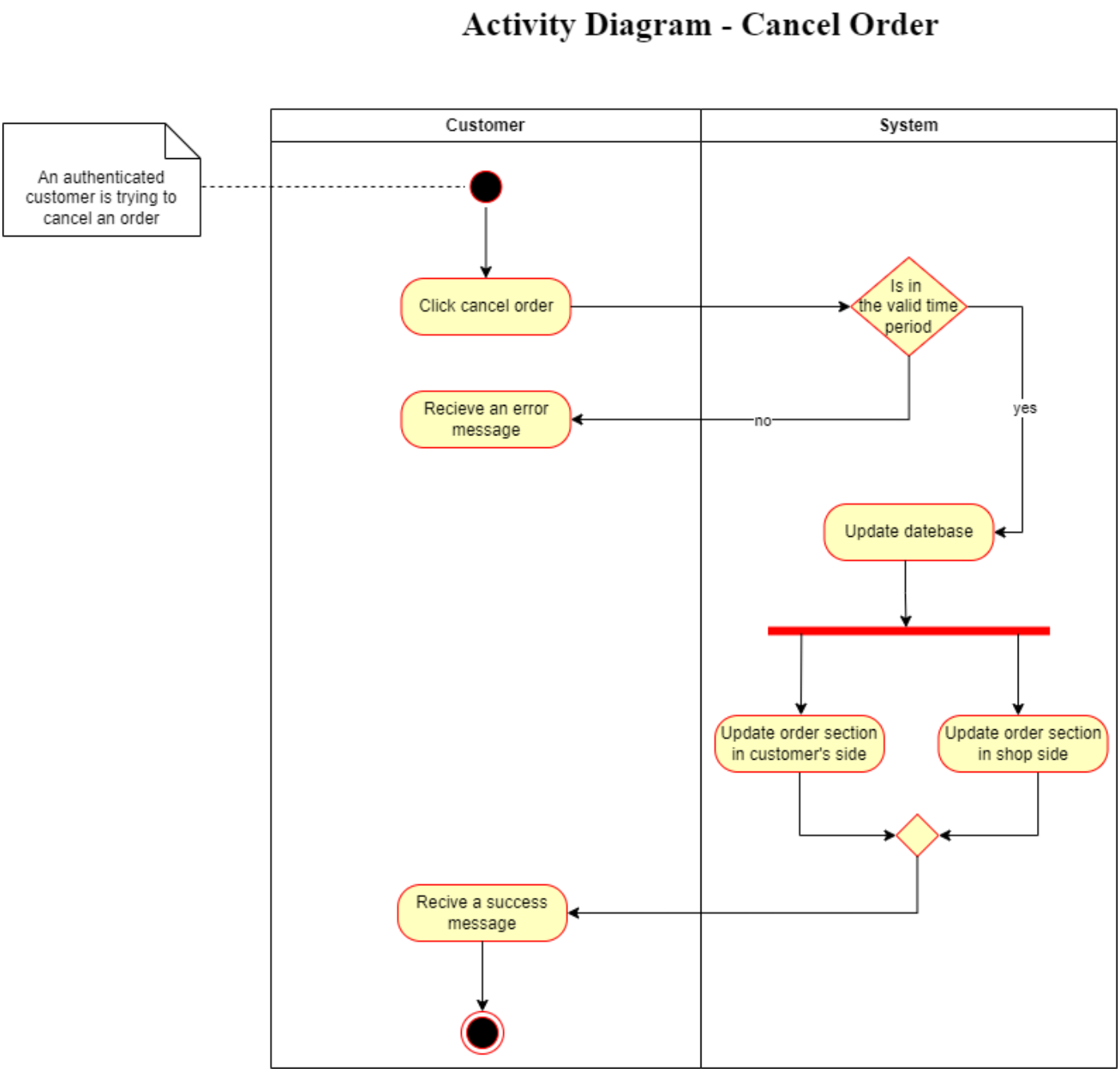

Figure 11 | Activity Diagram - Cancel Order

5.3.9 Order Tracking Activity

The user inputs the order ID to check the order status. The system retrieves real-time tracking information from the courier and displays it. If tracking is unavailable, a “Tracking Pending” message is shown, or if an incorrect ID is entered, the user is prompted to re-enter it.

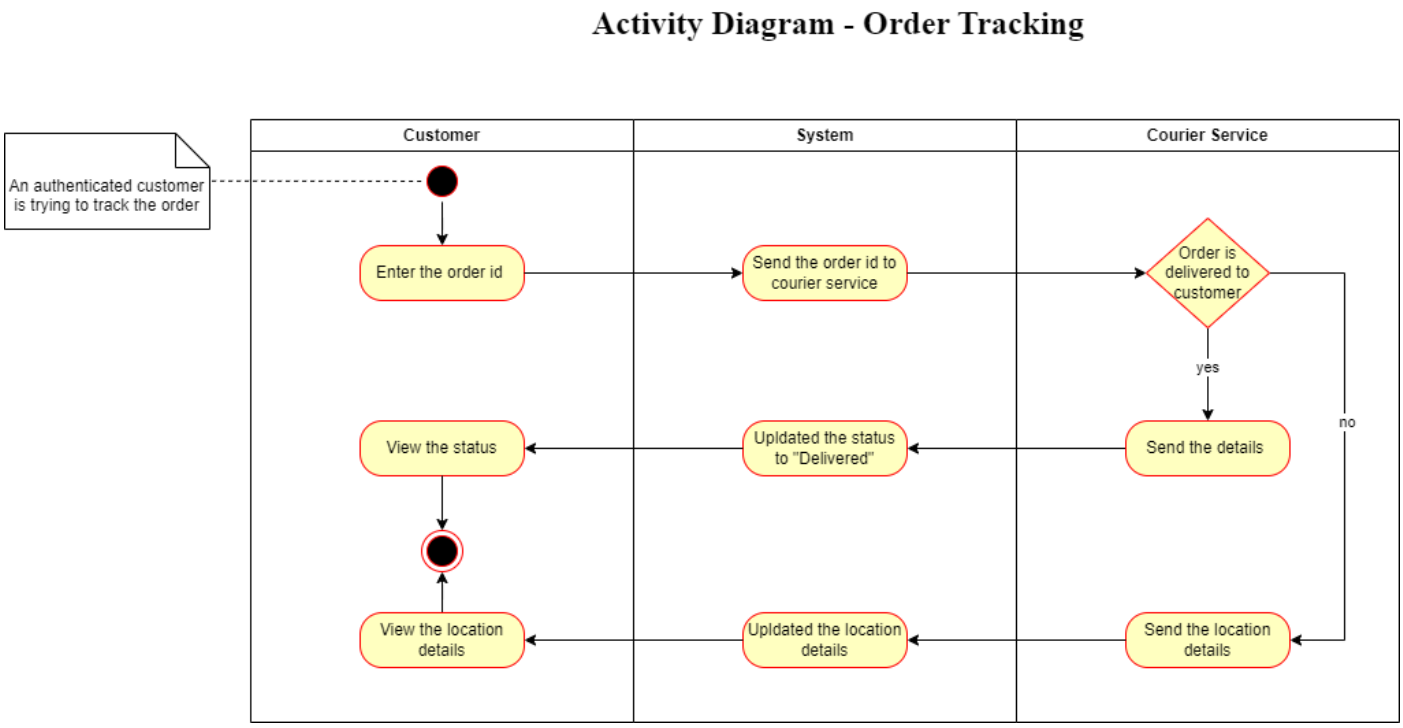

Figure 12 | Activity Diagram - Order Tracking

5.3.10 Return Order Management Activity

The user selects an order or item to return, initiating a return request. The system verifies eligibility, issues a return authorization, coordinates courier pickup, and updates the return status. If the return window has closed, the system notifies the user.

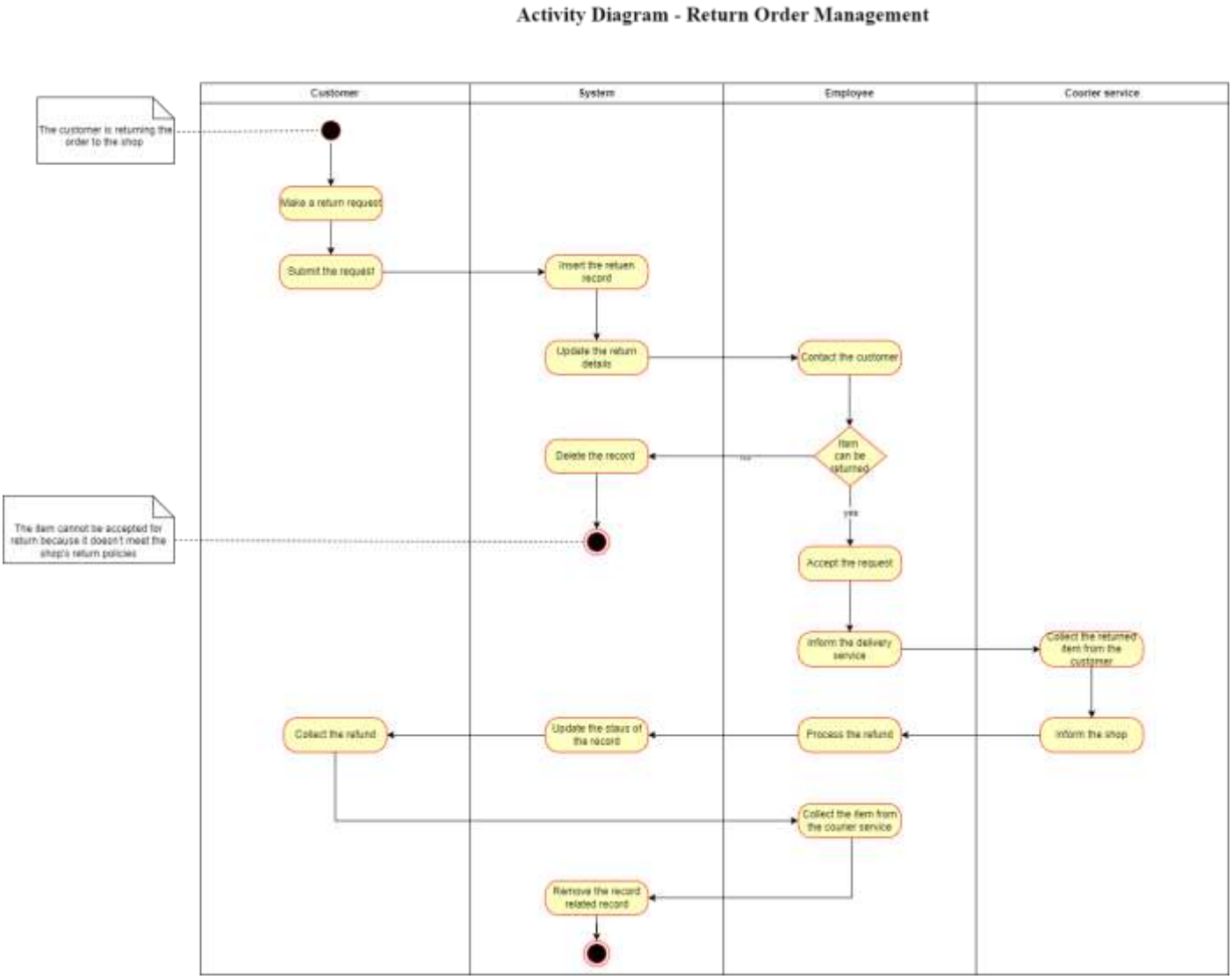

Figure 13 | Activity Diagram - Return Order Management

5.3.11 Manage Profile Activity

The user accesses profile settings to update personal information, and the system validates and saves changes. If required fields are incomplete or invalid, the system prompts for corrections. If saving fails, the user is notified and can retry.

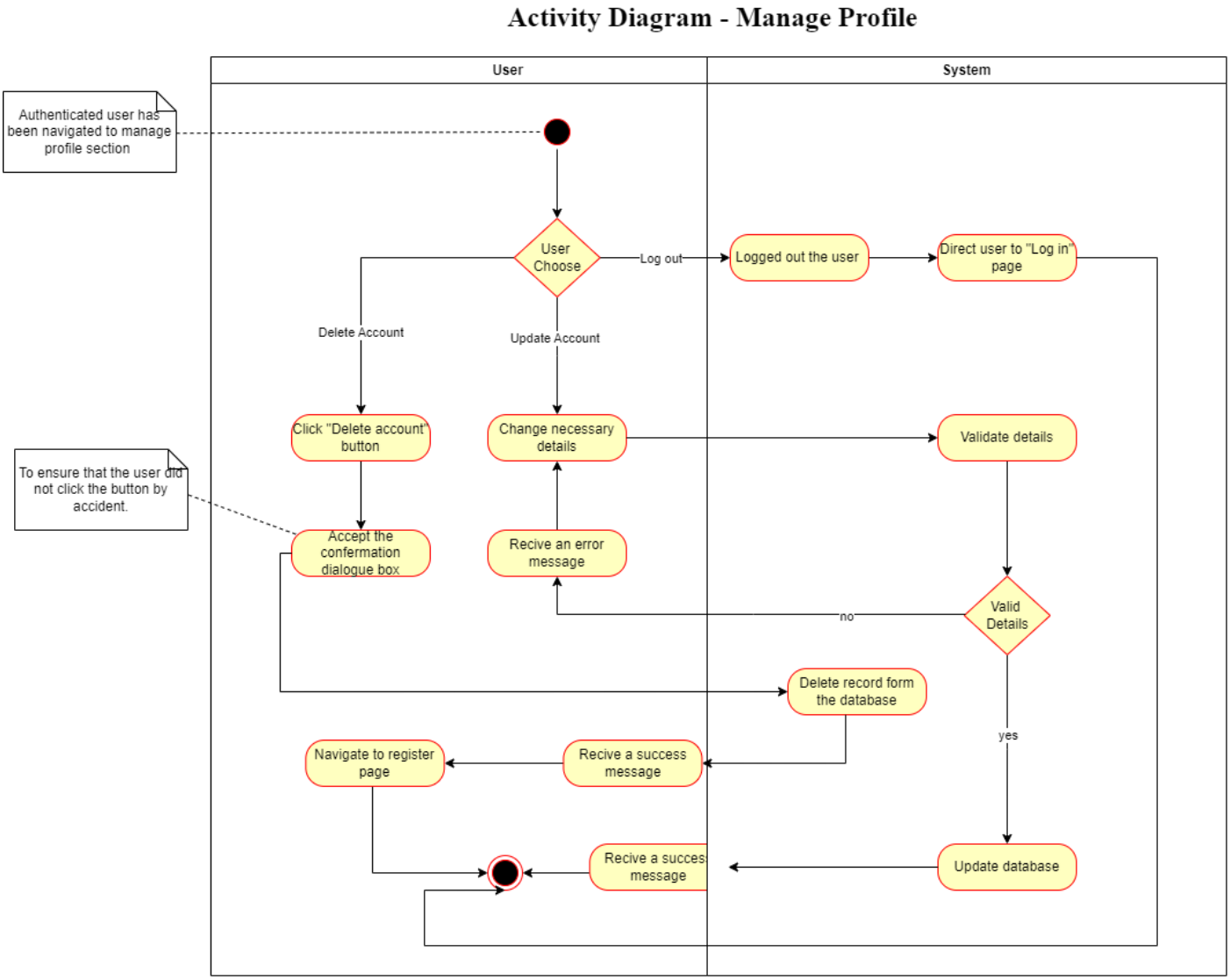

Figure 14 | Activity Diagram - Manage Profile

5.3.12 Add Complaint Activity

The user fills out and submits a complaint form. The system logs the complaint in the database and confirms the submission to the user. If the submission fails, the system displays an error message, and the user can retry.

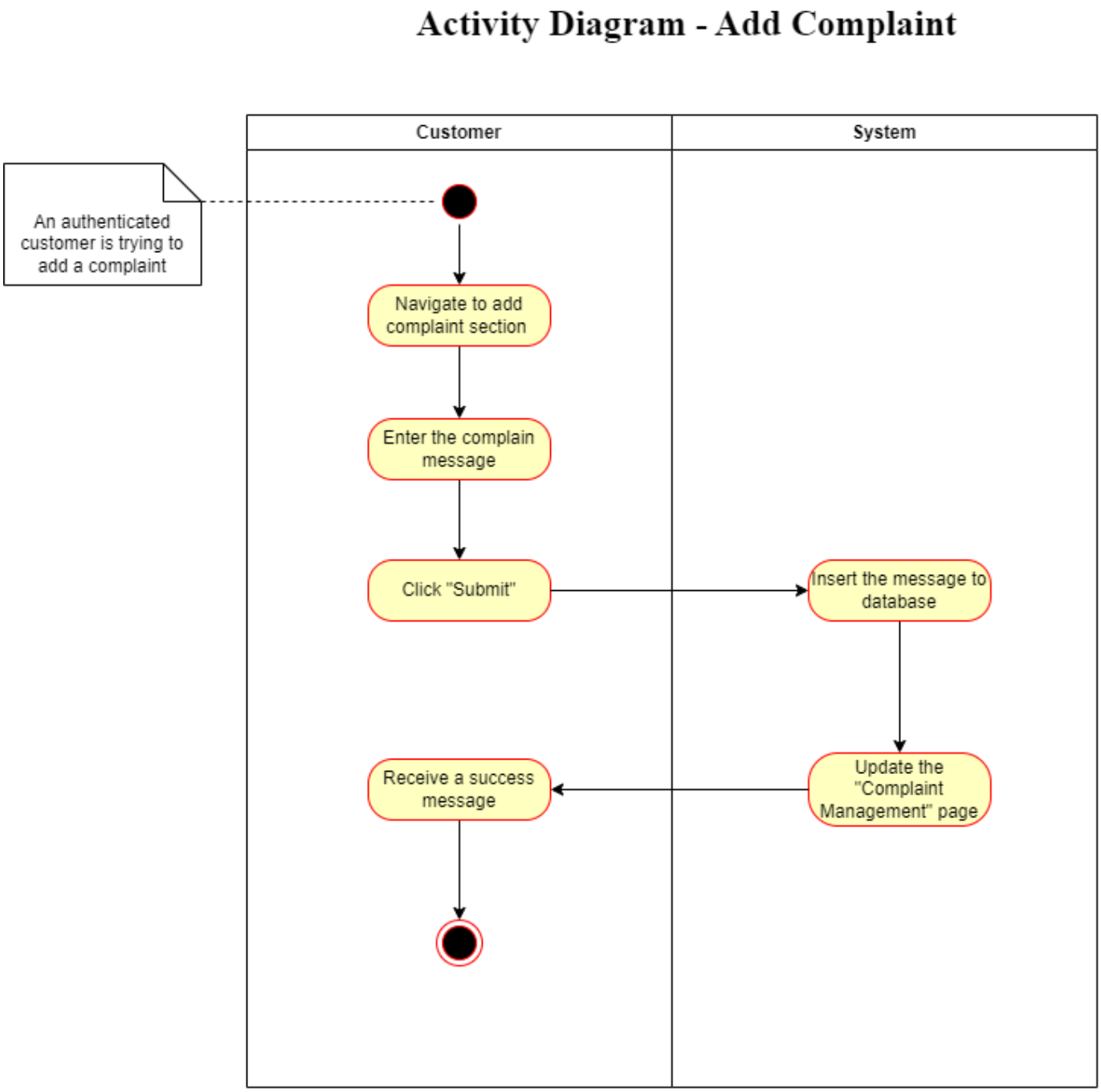

Figure 15 | Activity Diagram - Add Complaint

5.3.13 Manage Complaint Activity

Customer support accesses and reviews complaints, responding or escalating if needed. The system updates the complaint status and notifies the user. If escalation is necessary, the support team forwards the complaint, with the system updating the status.

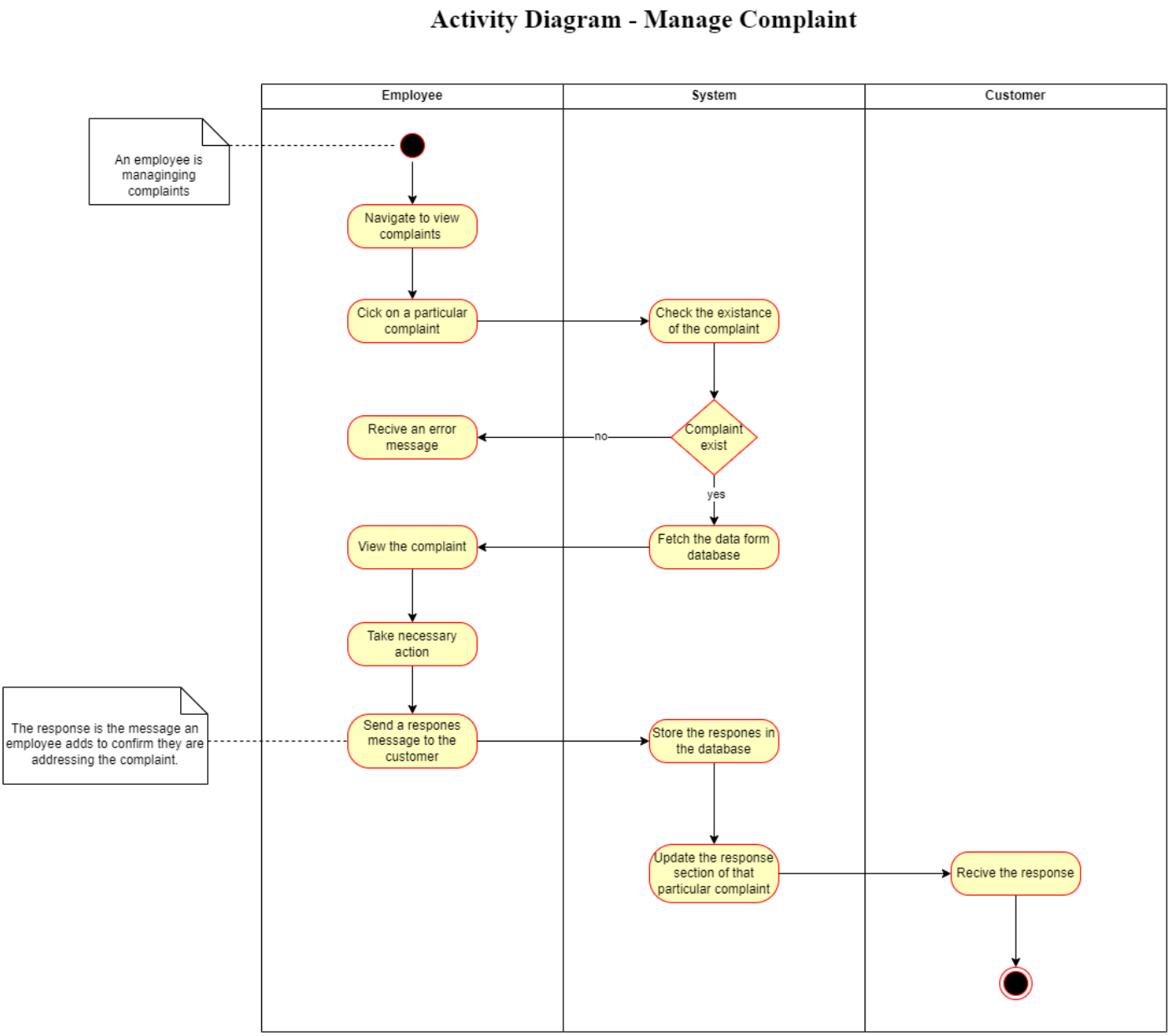

Figure 16 | Activity Diagram - Manage Complaint

5.3.14 Add Item Activity

The admin or vendor accesses the inventory section, enters new item details, and submits them. The system validates and adds the product to the catalogue. If mandatory details are missing, the system prompts for completion, and errors are displayed if the addition fails.

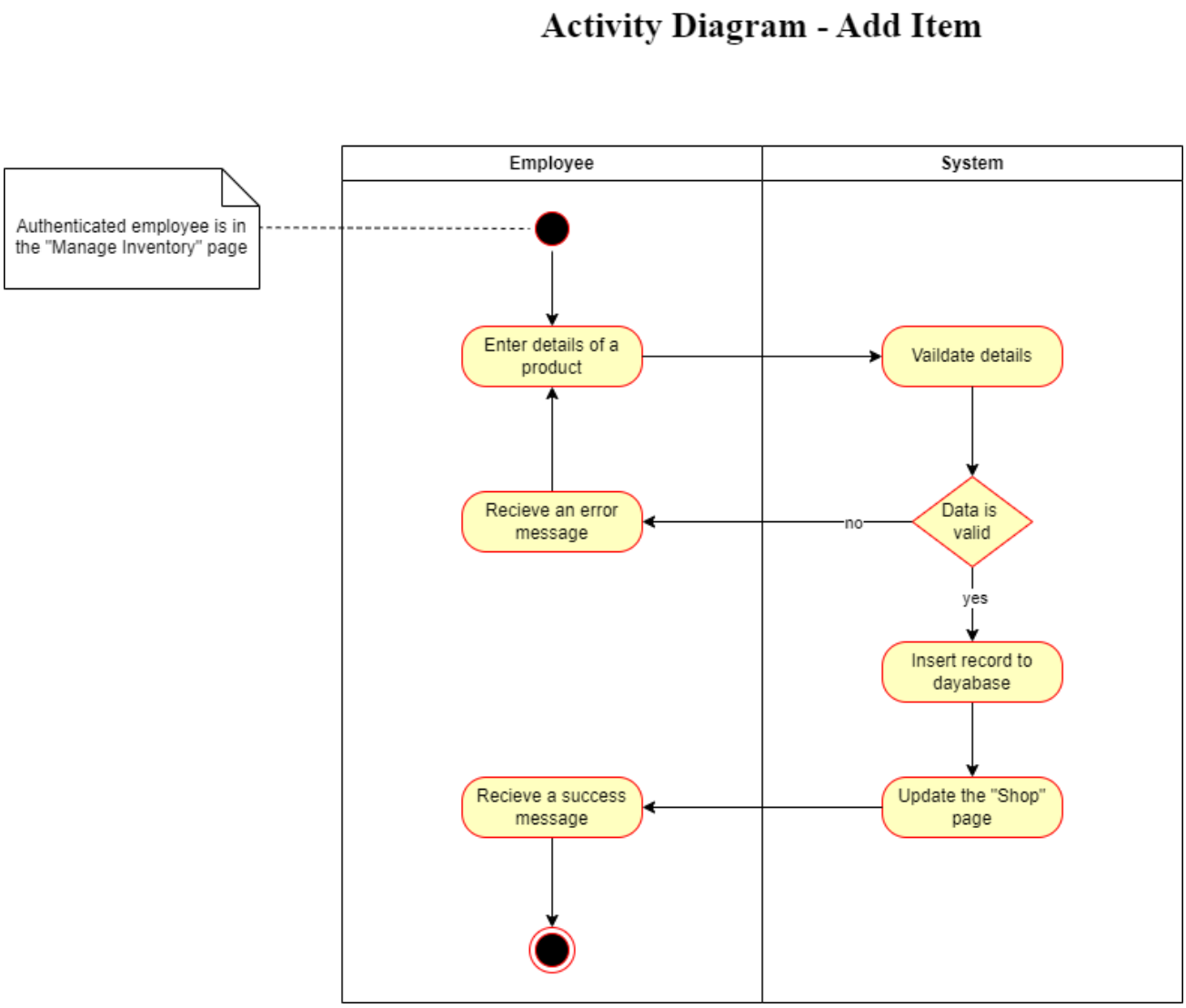

Figure 17 | Activity Diagram - Add Item

5.3.15 Update Item Activity

Admin or vendor selects an item, updates details, and saves. The system validates and updates the item in the catalogue, confirming changes to the vendor. If validation fails, the vendor is prompted to correct details, and errors display if updating fails.

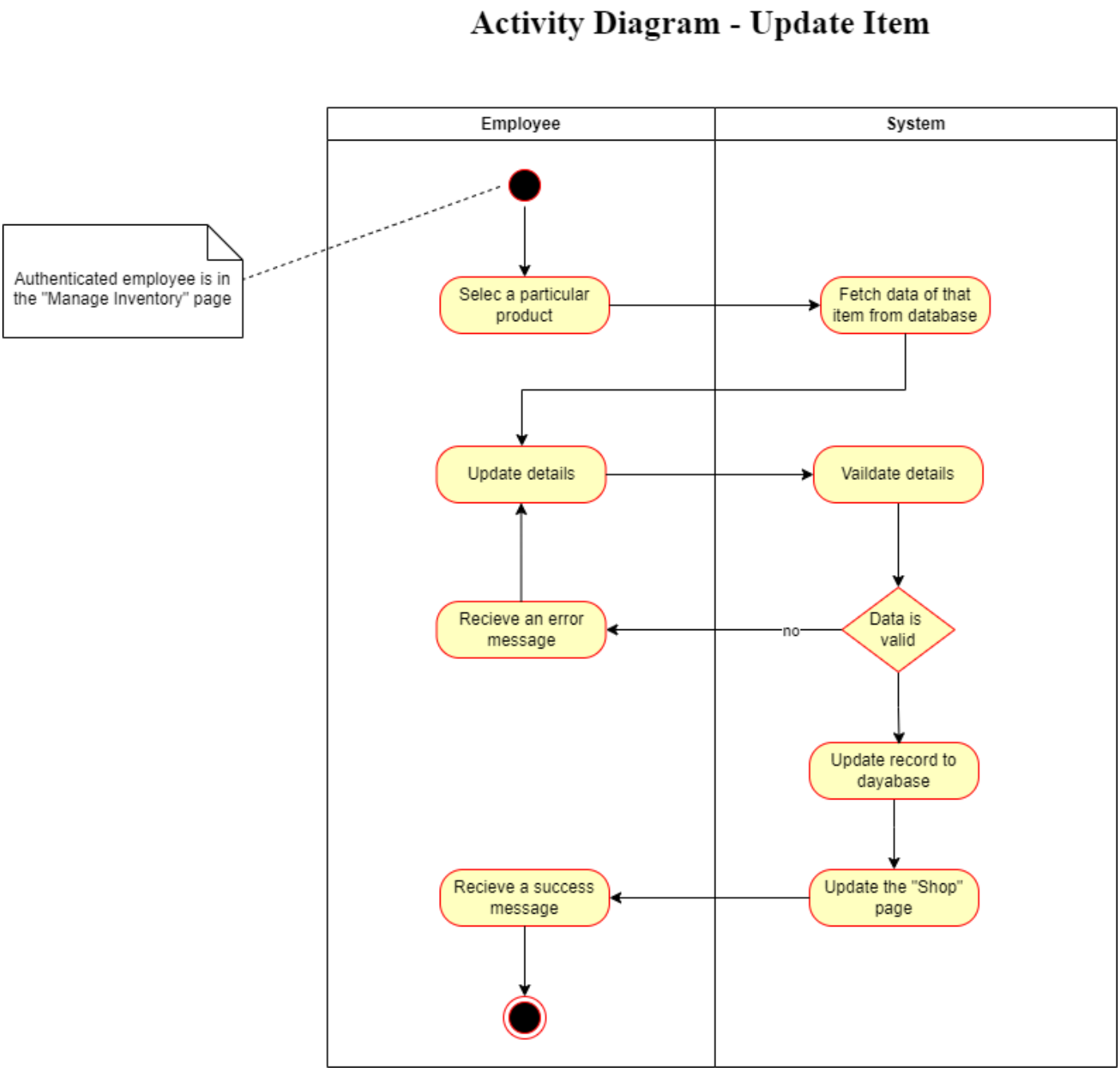

Figure 18 | Activity Diagram - Update Item

5.3.16 Delete Item Activity

The admin or vendor selects an item to delete, confirms, and submits it. The system checks dependencies and, if none exist, removes the item from the catalogue. Deletion is denied if active orders are linked to the item, and the system notifies the admin.

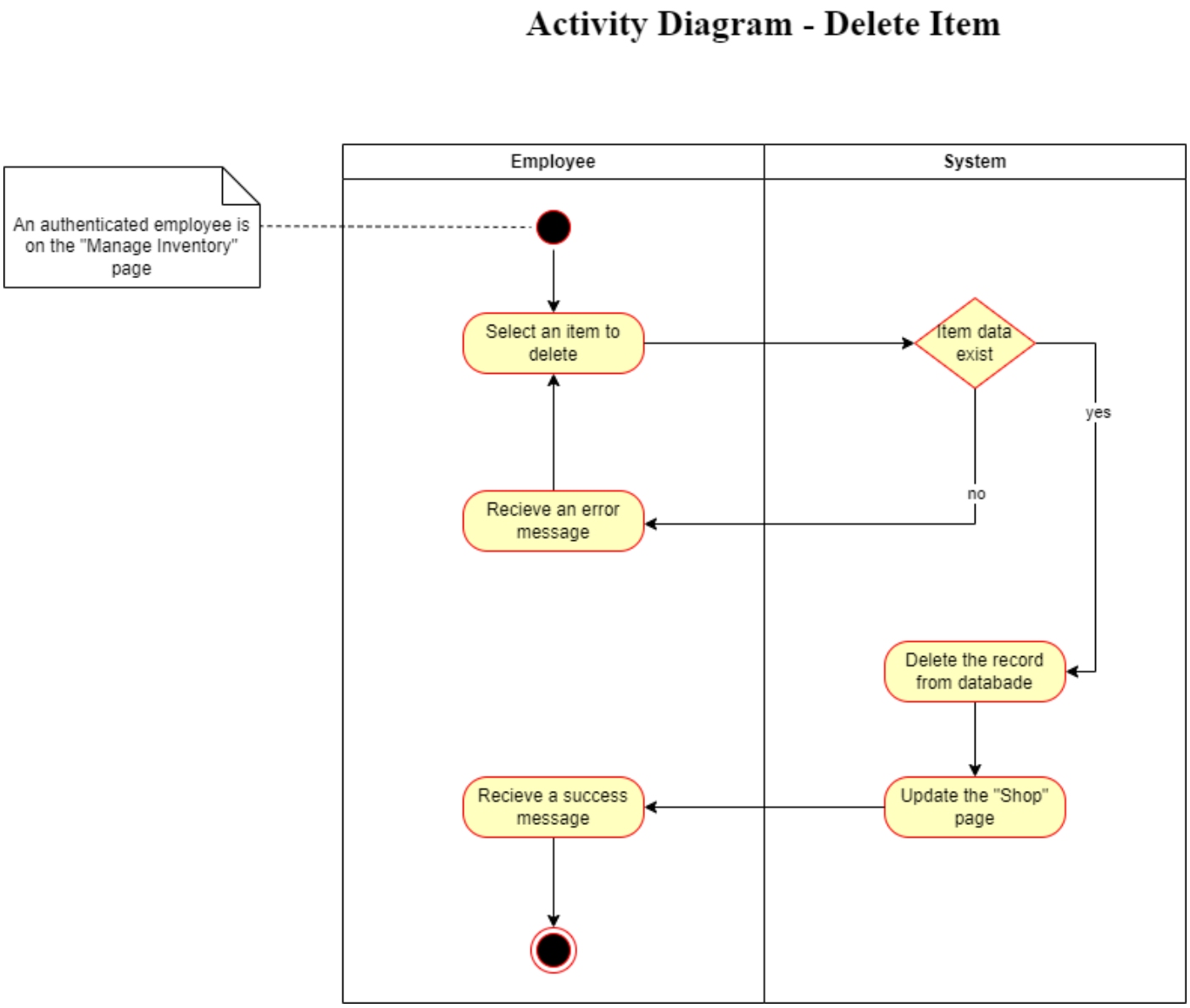

Figure 19 | Activity Diagram - Delete Item

5.4 Class Diagrams

A class diagram is a type of UML (Unified Modelling Language) diagram used to show the relationships and structure of classes in a system. Class diagrams help with visualization because they give stakeholders a visual depiction of the system's structure, making it simpler to comprehend the dependencies and relationships between classes. Three class diagrams—the interface, controller, and entity—are created based on stereotypes to provide further insight.

5.4.1 Entity Class

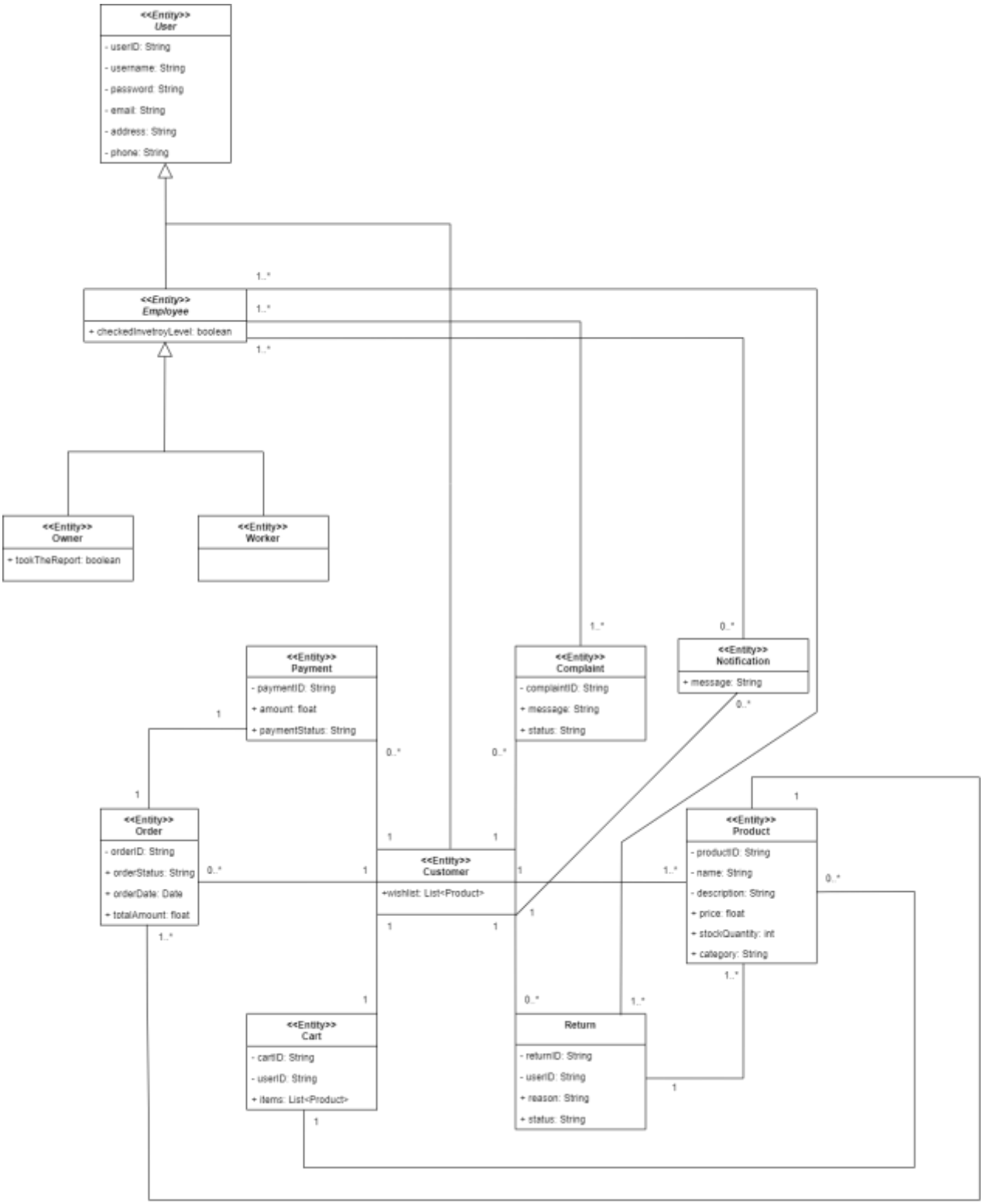

Figure 20 | Entity Class

5.4.2 Controller Class

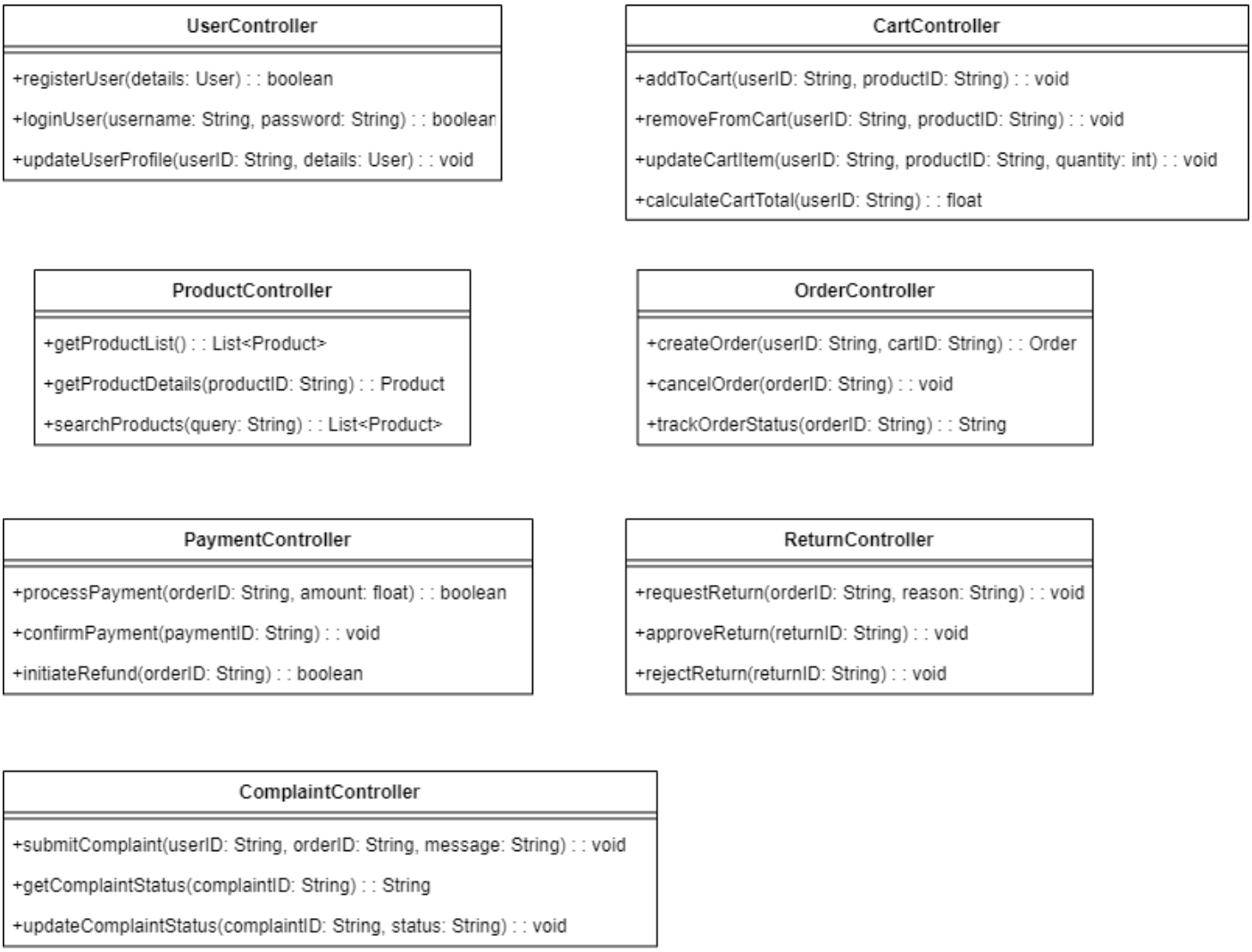

Figure 21 | Controller Class

5.4.3 Interface Class

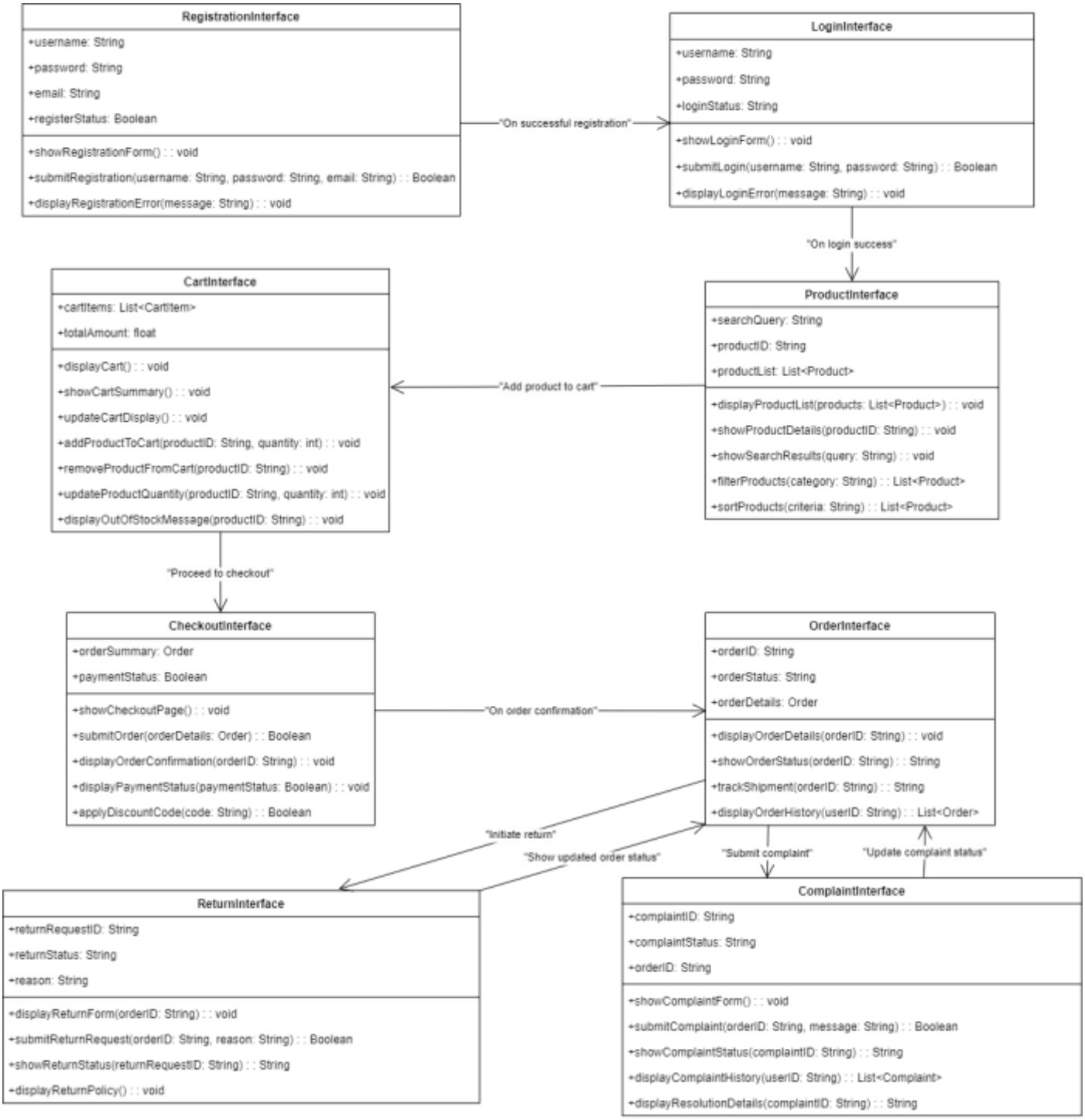

Figure 22 | Interface Class

5.5 Sequence Diagrams

Sequence diagrams, a type of UML diagram, are used to illustrate how different components or objects within a system interact and communicate with one another. They provide a dynamic view of the system by emphasizing the flow of control and the chronological order of interactions.

Understanding the dynamic behaviour of a system is aided by sequence diagrams, which illustrate the order in which interactions between objects or components occur.

5.5.1 User Registration Sequence Diagram

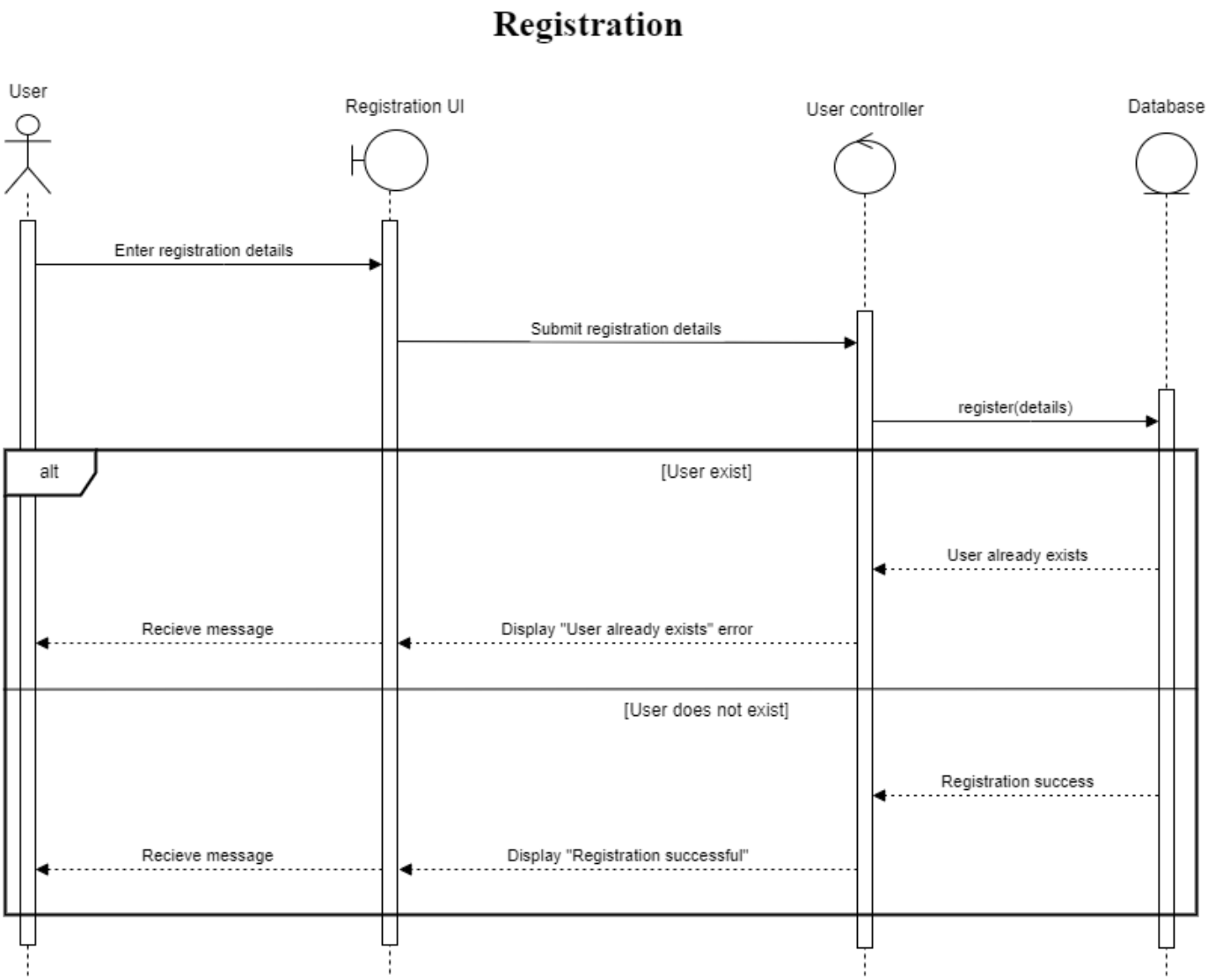

Figure 23 | Sequence diagram - User Registration

5.5.2 User Login Sequence Diagram

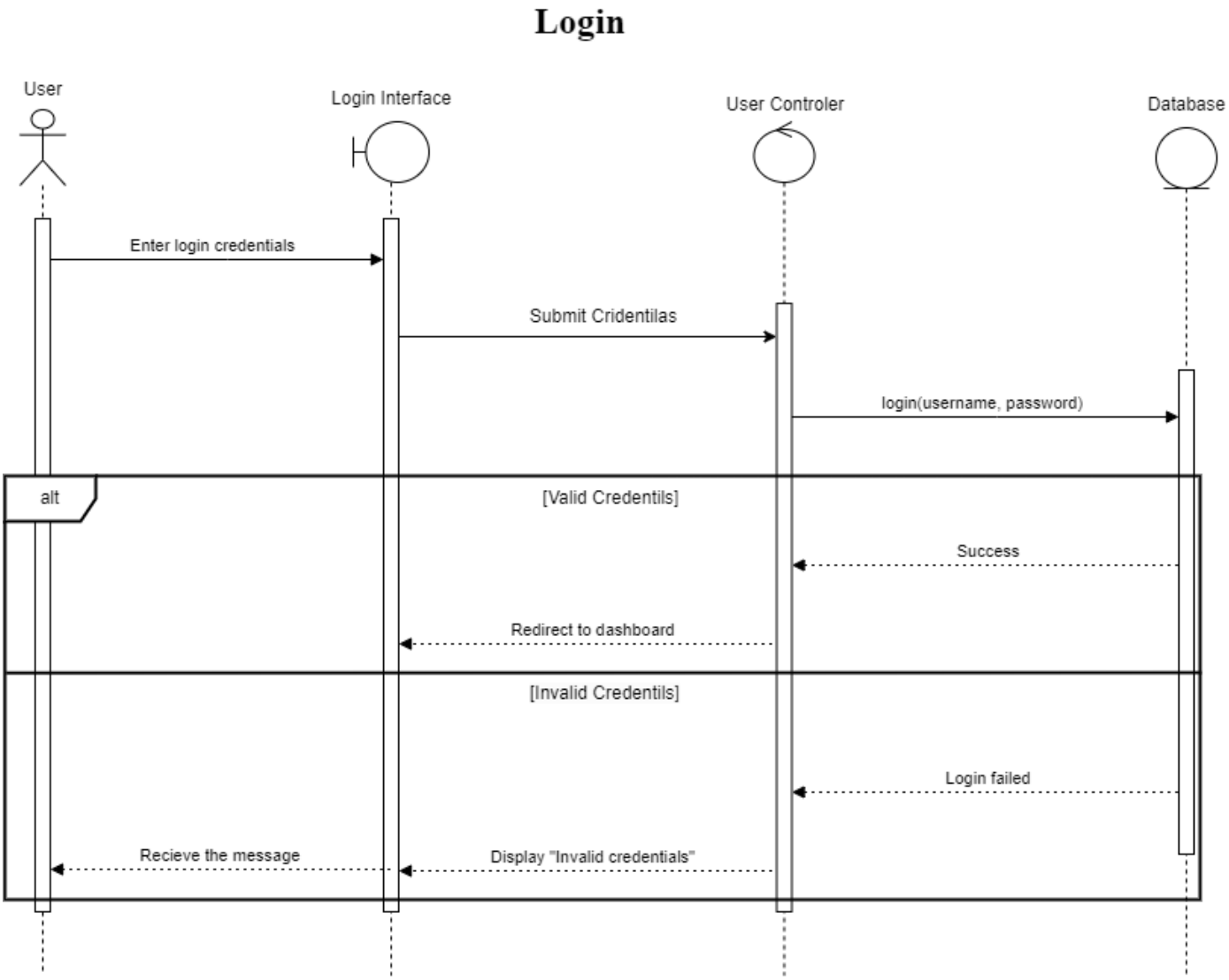

Figure 24 | Sequence diagram - User Log In

5.5.3 Product Search Sequence Diagram

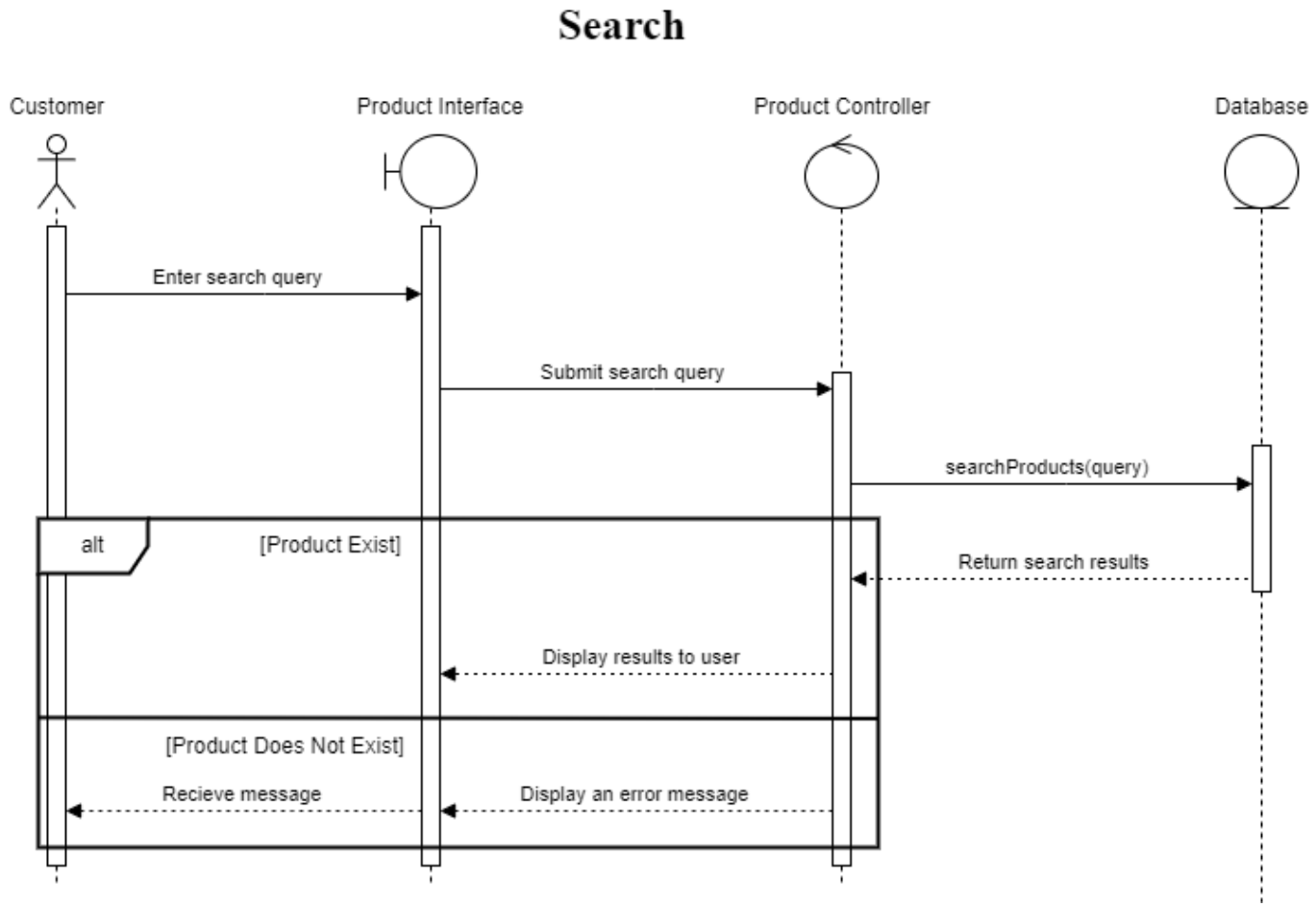

Figure 25 | Sequence diagram - Search

5.5.4 Add to Cart Sequence Diagram

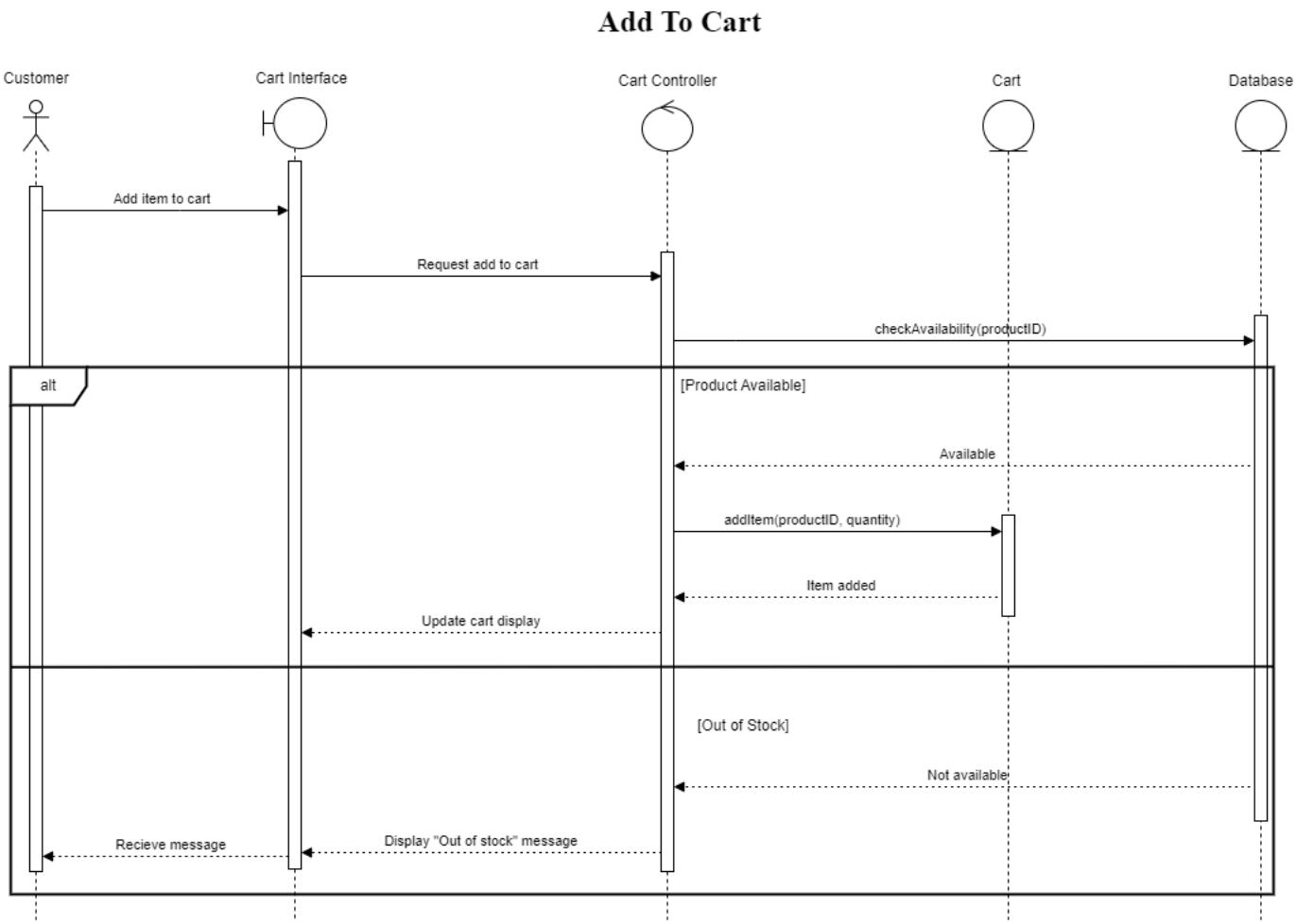

Figure 26 | Sequence diagram - Add to Cart

5.5.5 Place Order Sequence Diagram

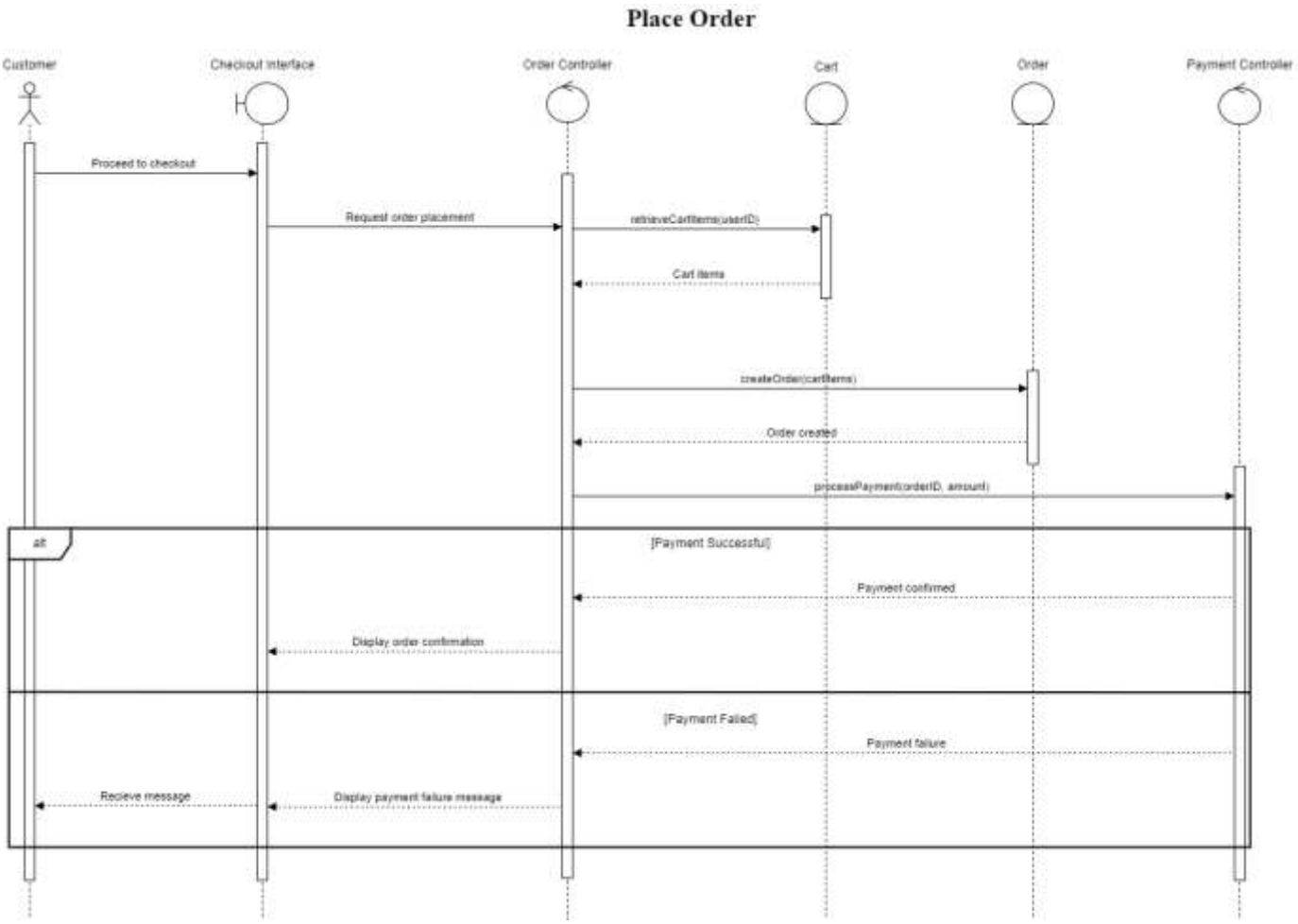

Figure 27 | Sequence diagram - Place Order

5.5.6 Order Tracking Sequence Diagram

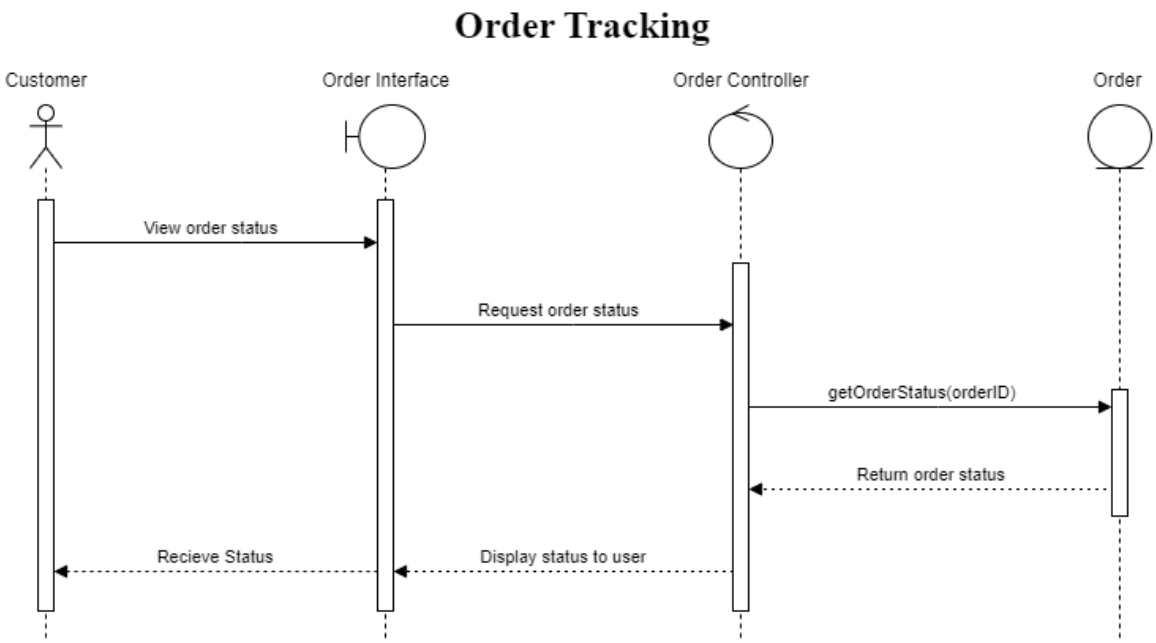

Figure 28 | Sequence diagram - Order Tracking

5.5.7 Return Order Sequence Diagram

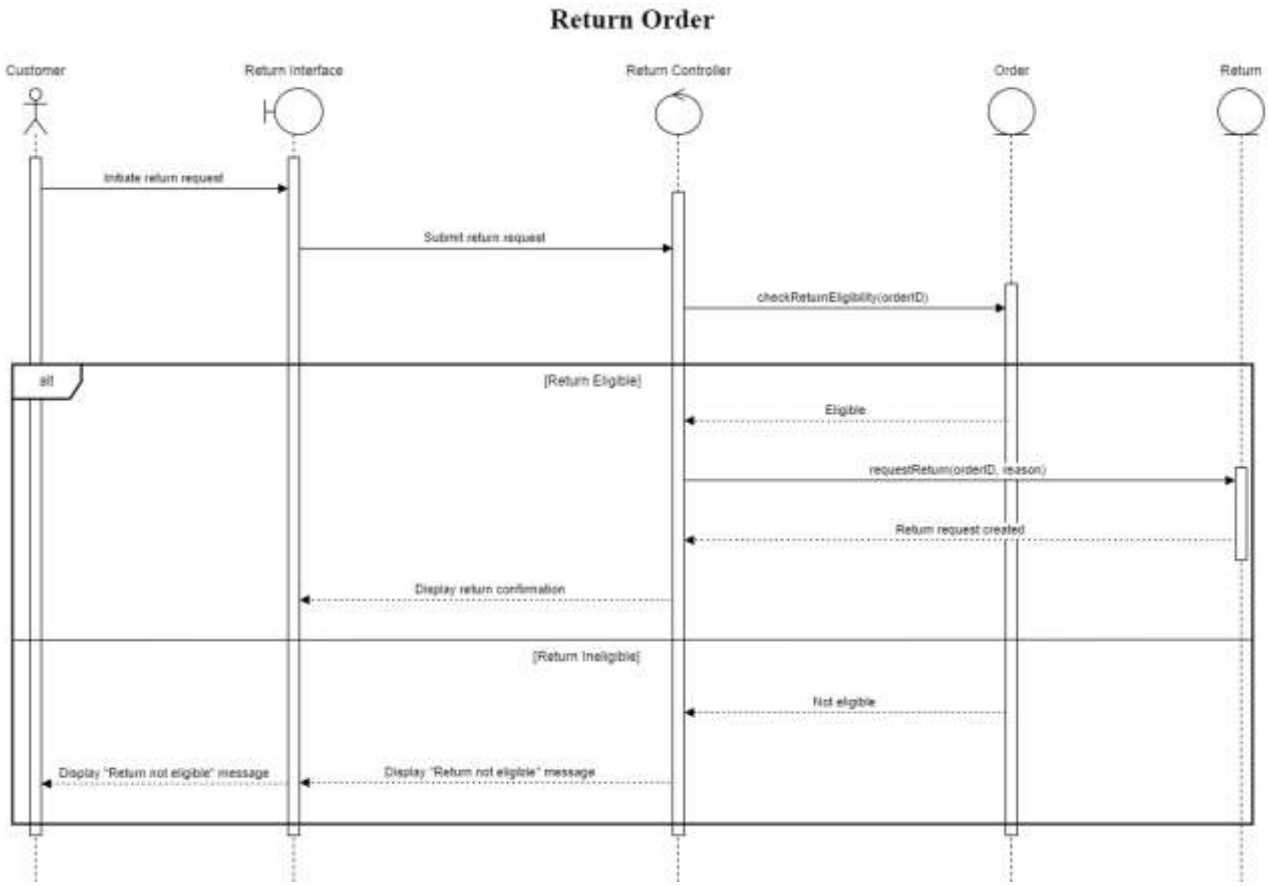

Figure 29 | Sequence diagram - Return Order

5.5.8 Submit Complaint Sequence Diagram

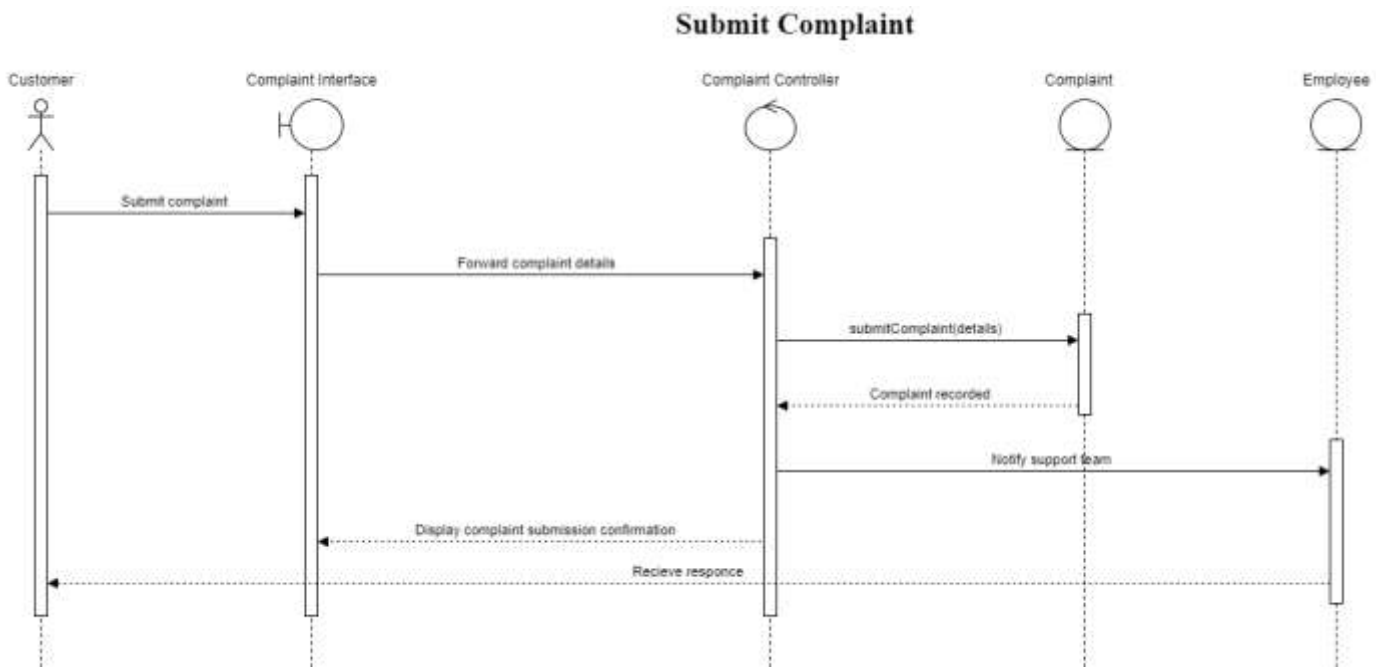

Figure 30 | Sequence diagram - Submit Complaint

5.6 Normalized Data Model

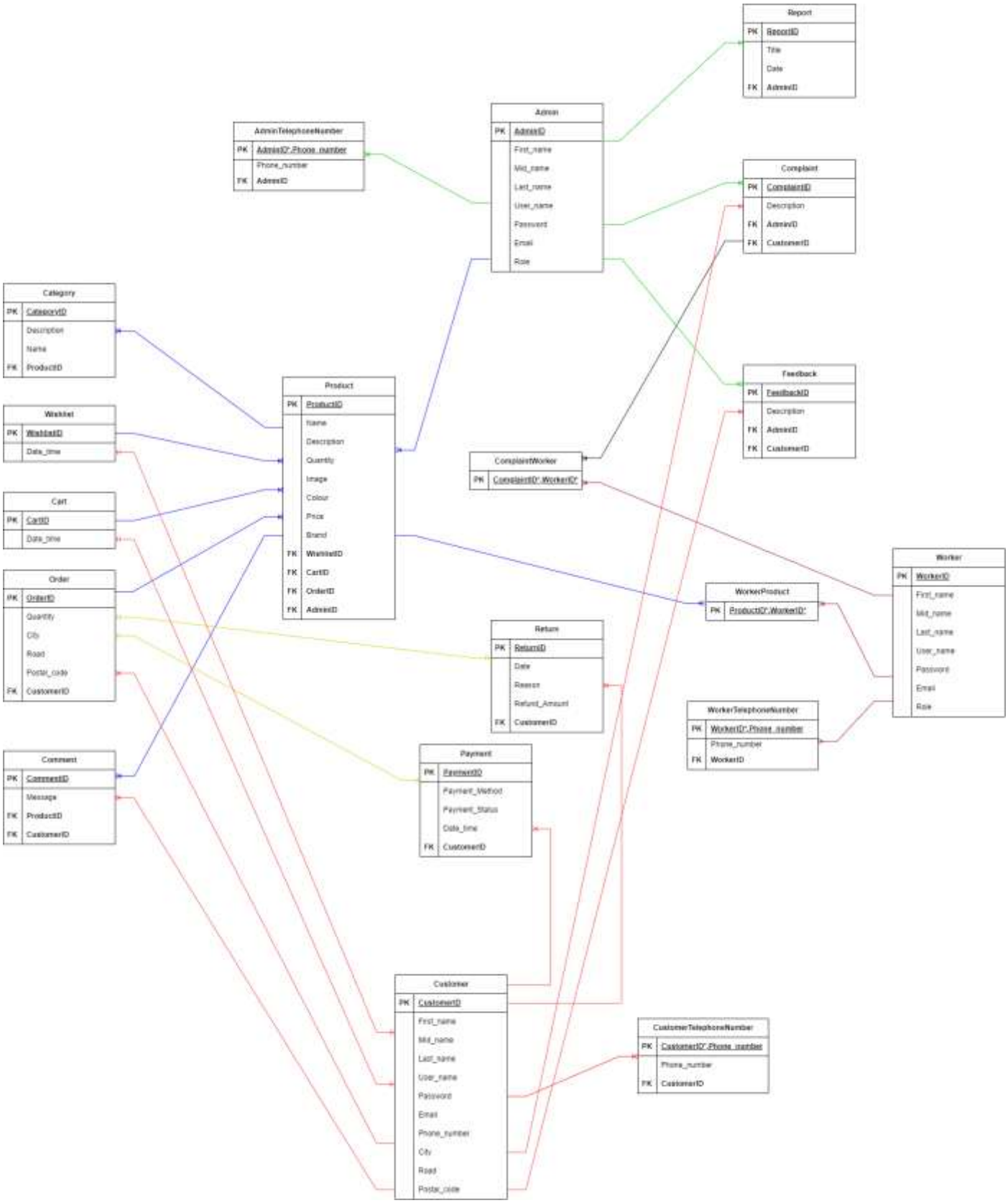

Figure 31 | Normalized Data Model

5.7 ER Diagram

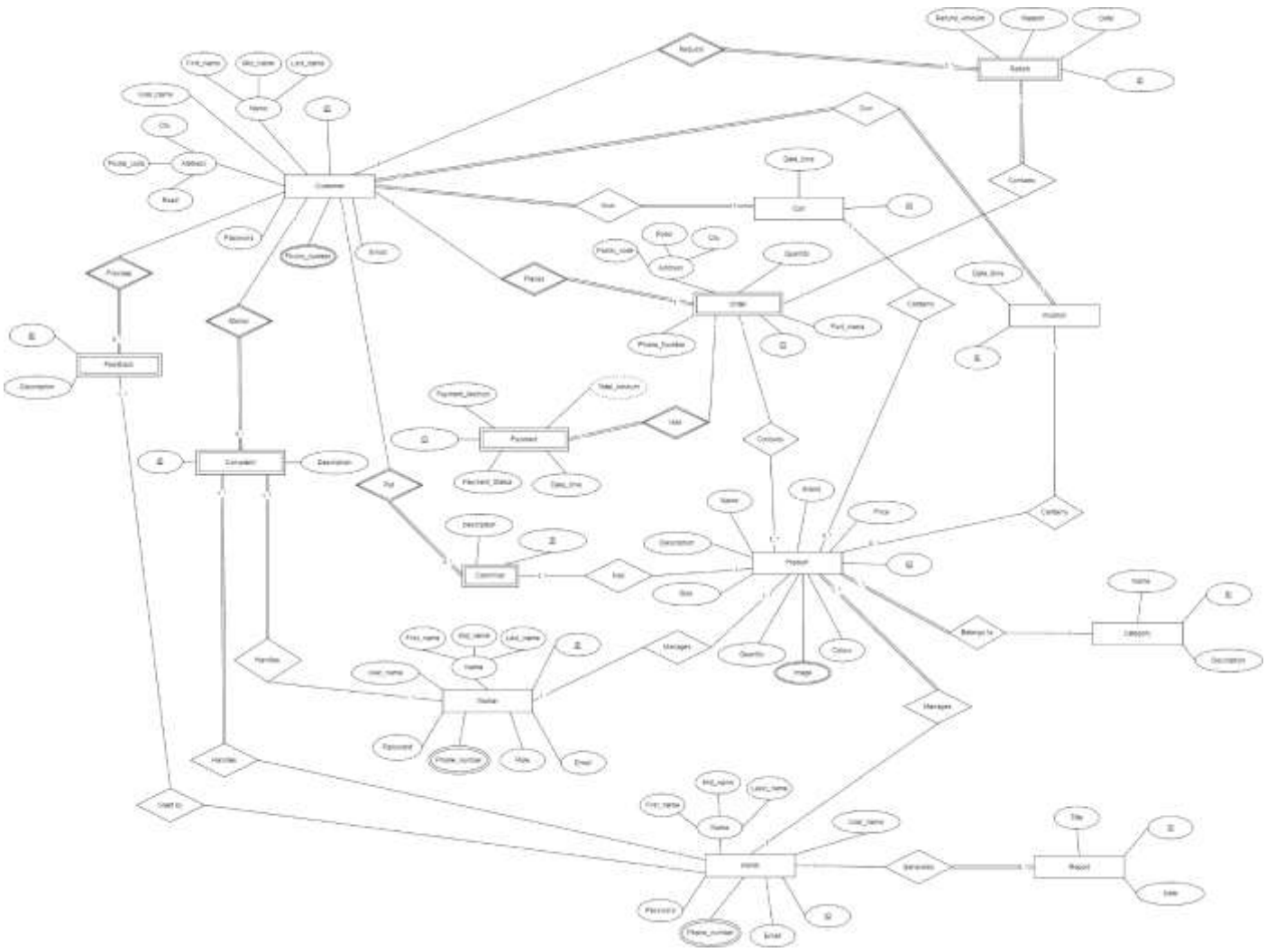

Figure 32 | ER Diagram

5.8 Graphical User Interfaces

The User Interface (UI) of the CMax Online Store web application is designed to provide a seamless and intuitive user experience. It is crafted with careful consideration of user needs, preferences, and accessibility standards. The UI is built using Figma, which allows for a visually appealing and responsive design.

5.8.1 Customer Side

5.8.1.1 Sign Up

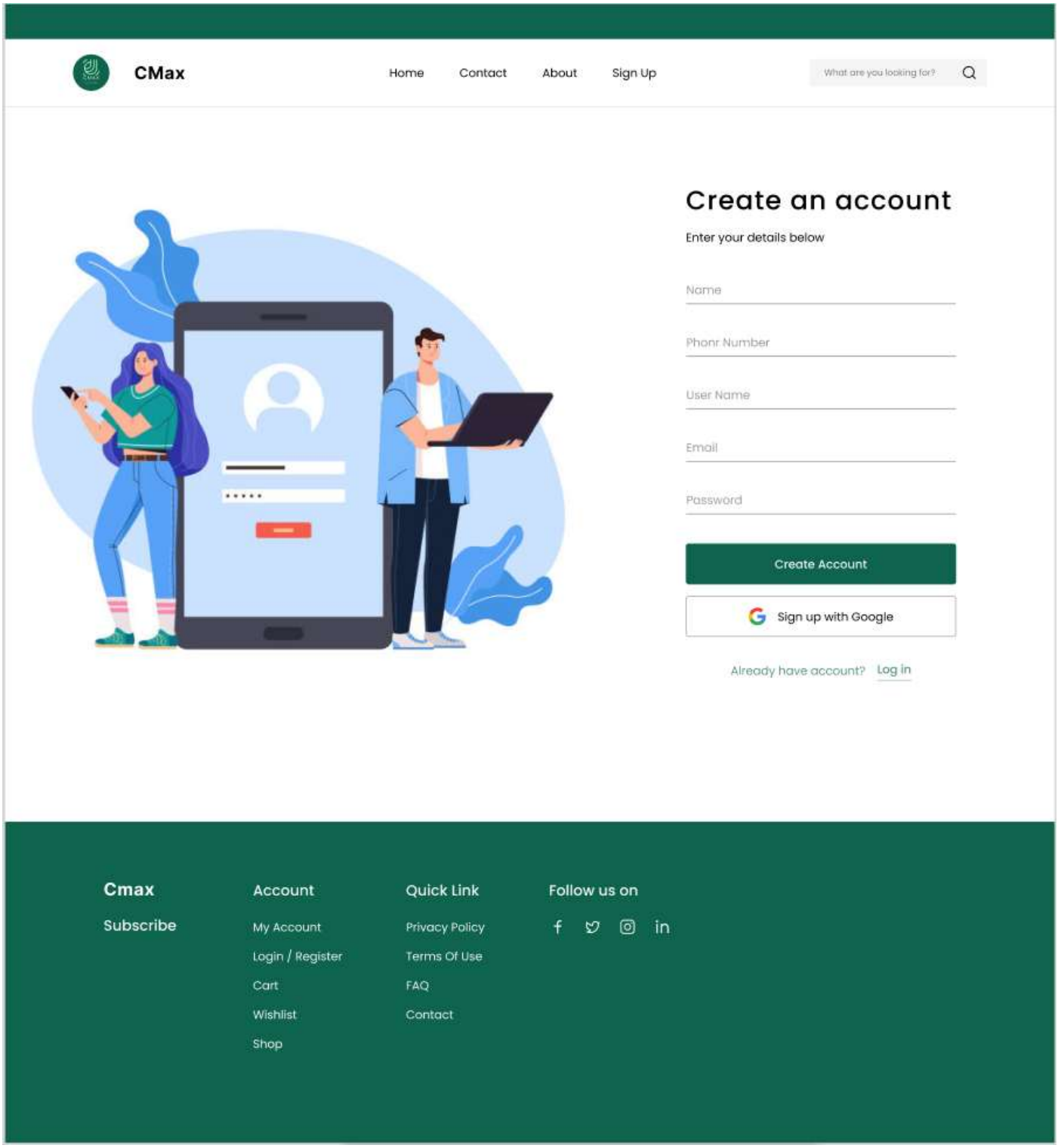

Figure 33 | Sign Up Page

5.8.1.2 Log In

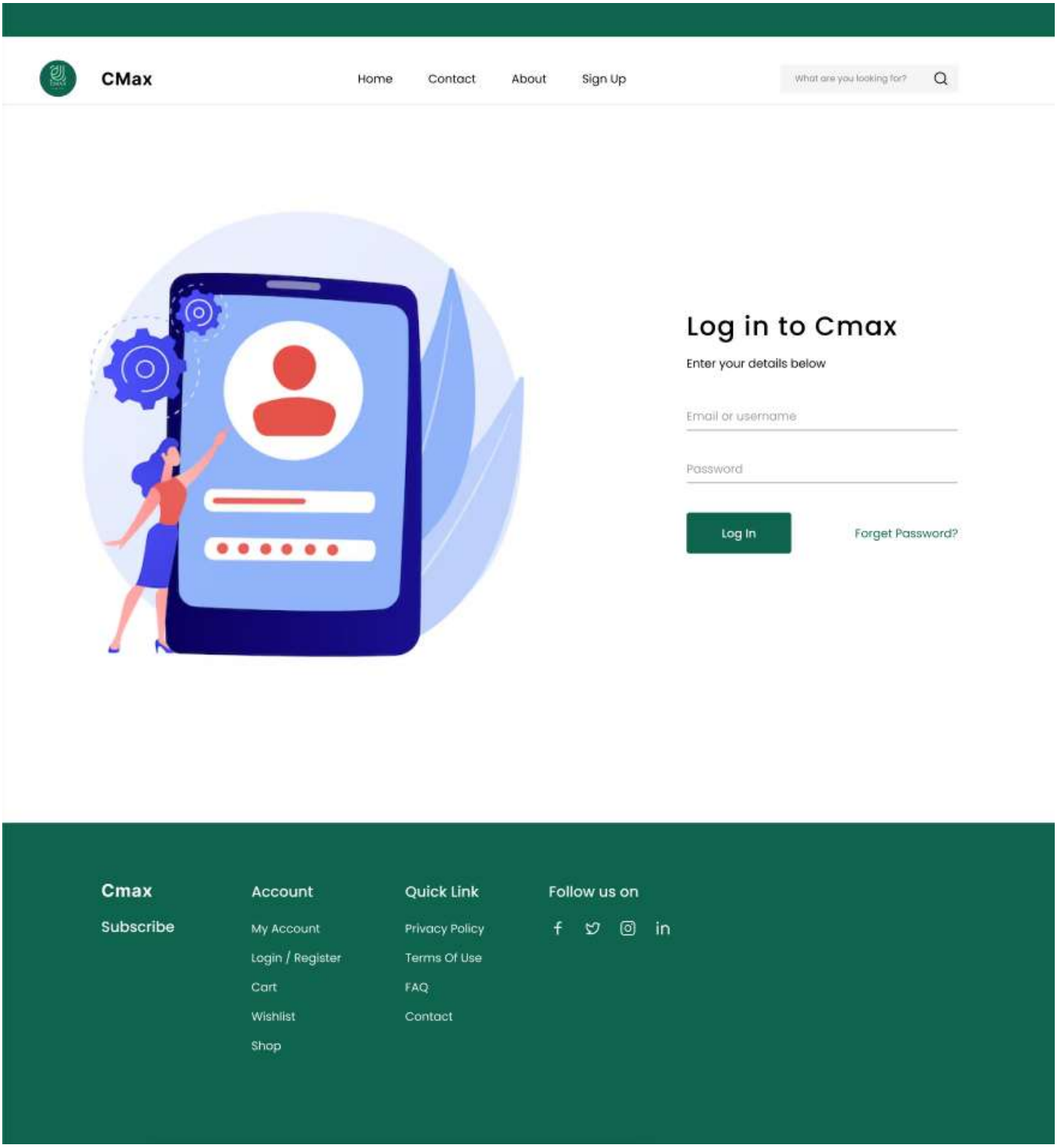

Figure 34 | Log in Page

5.8.1.3 Home

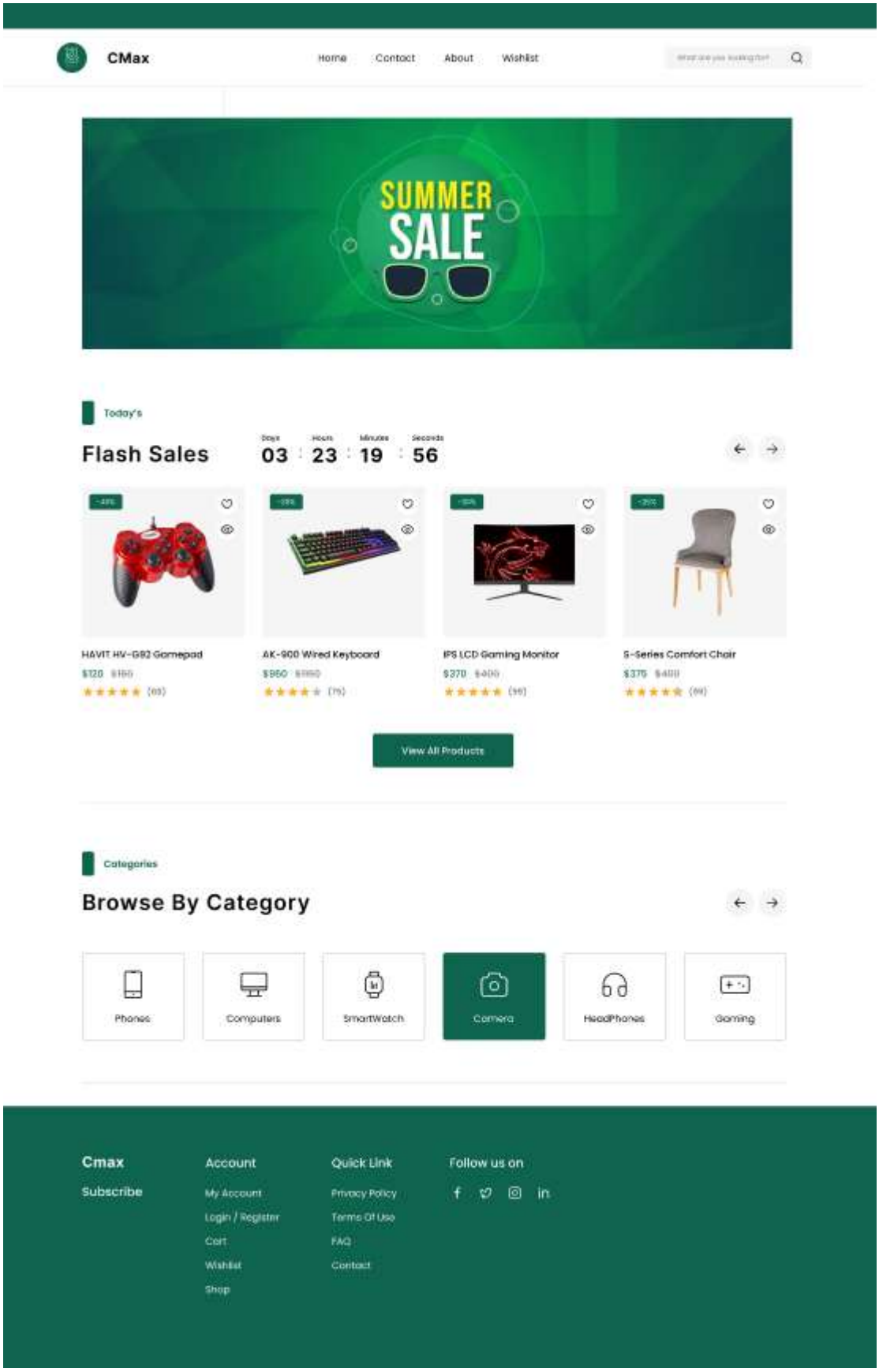

Figure 35 | Home Page

5.8.1.4 View Product

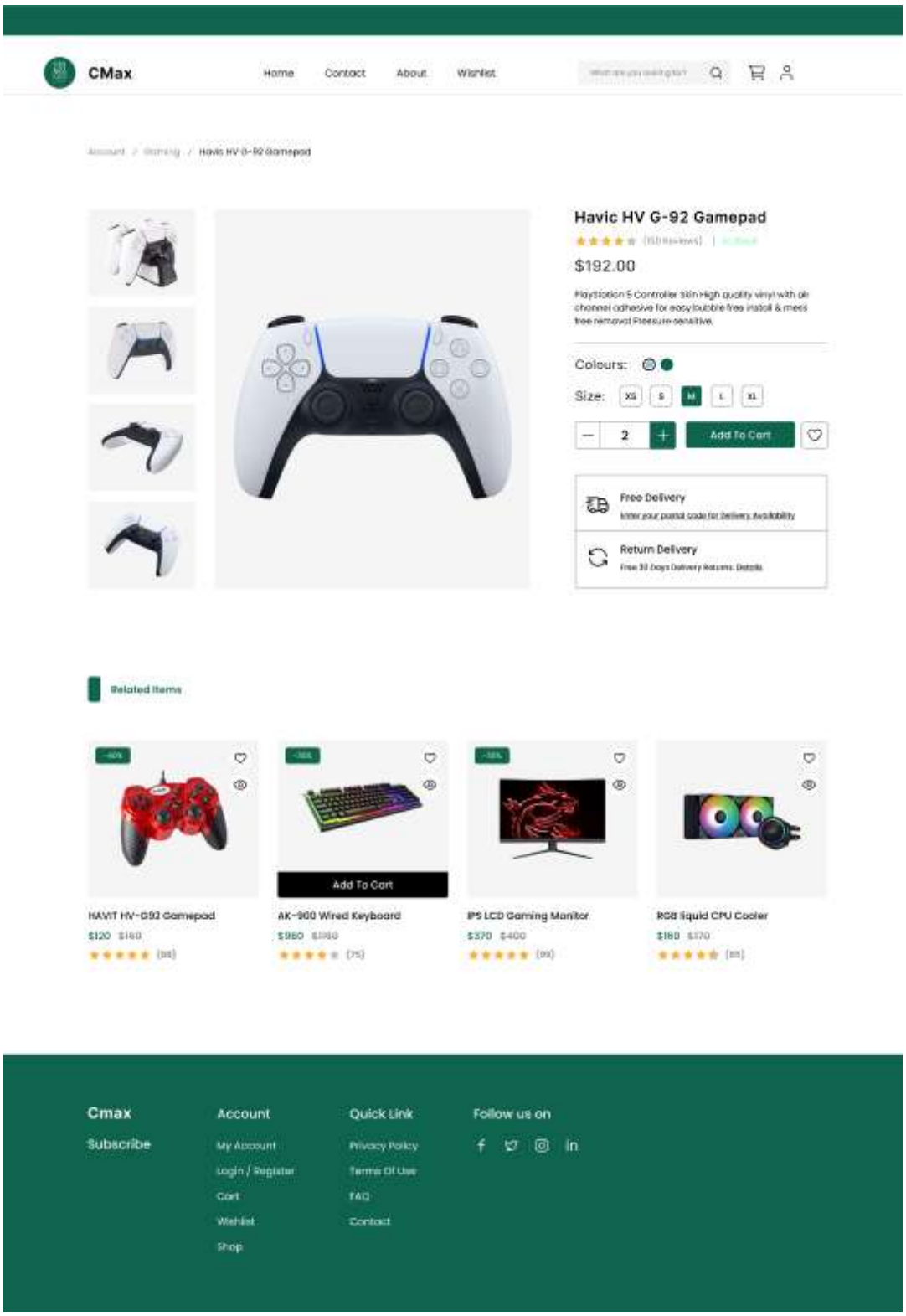

Figure 36 | View Product Page

5.8.1.5 Wishlist

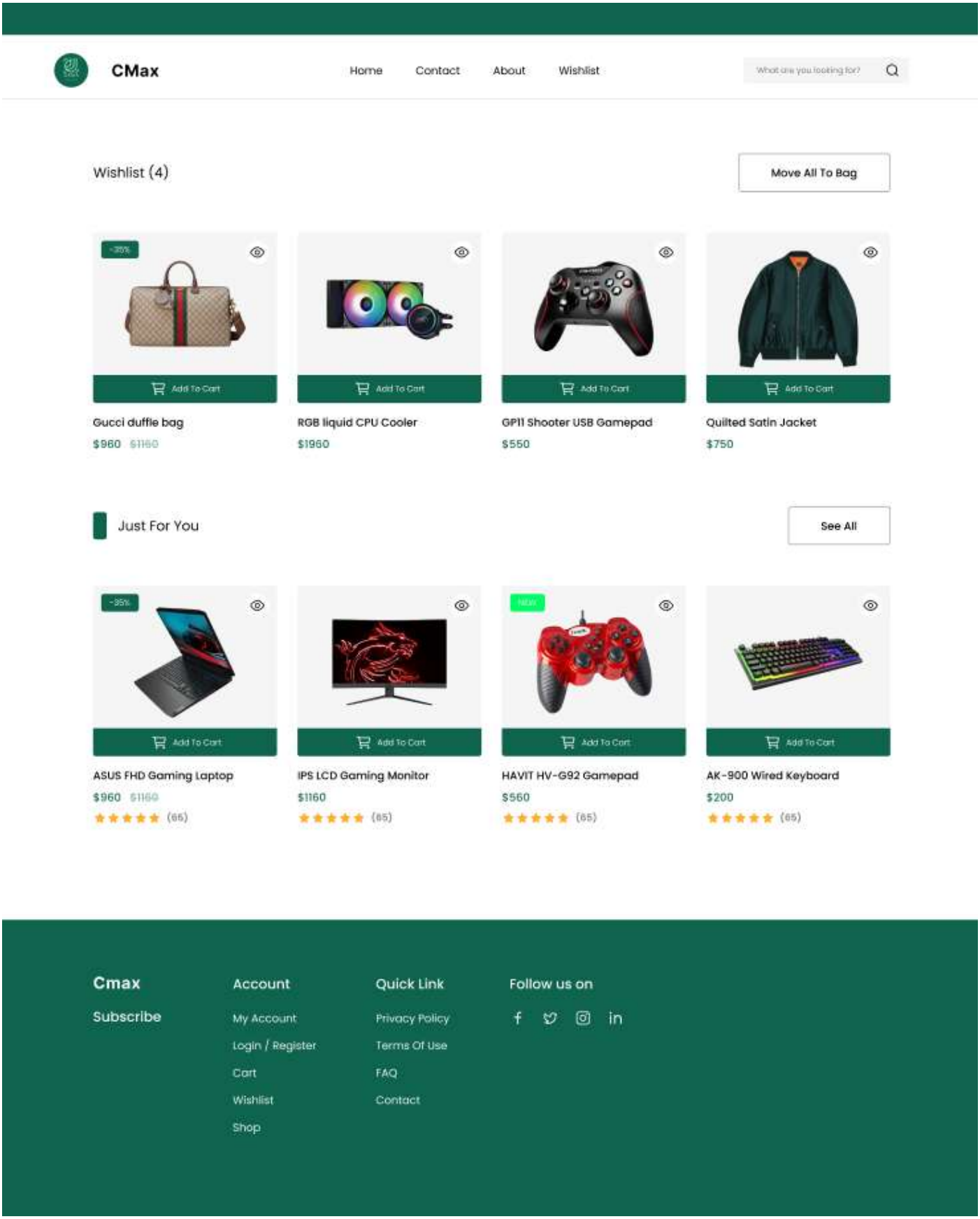

Figure 37 | Wishlist Page

5.8.1.6 Cart

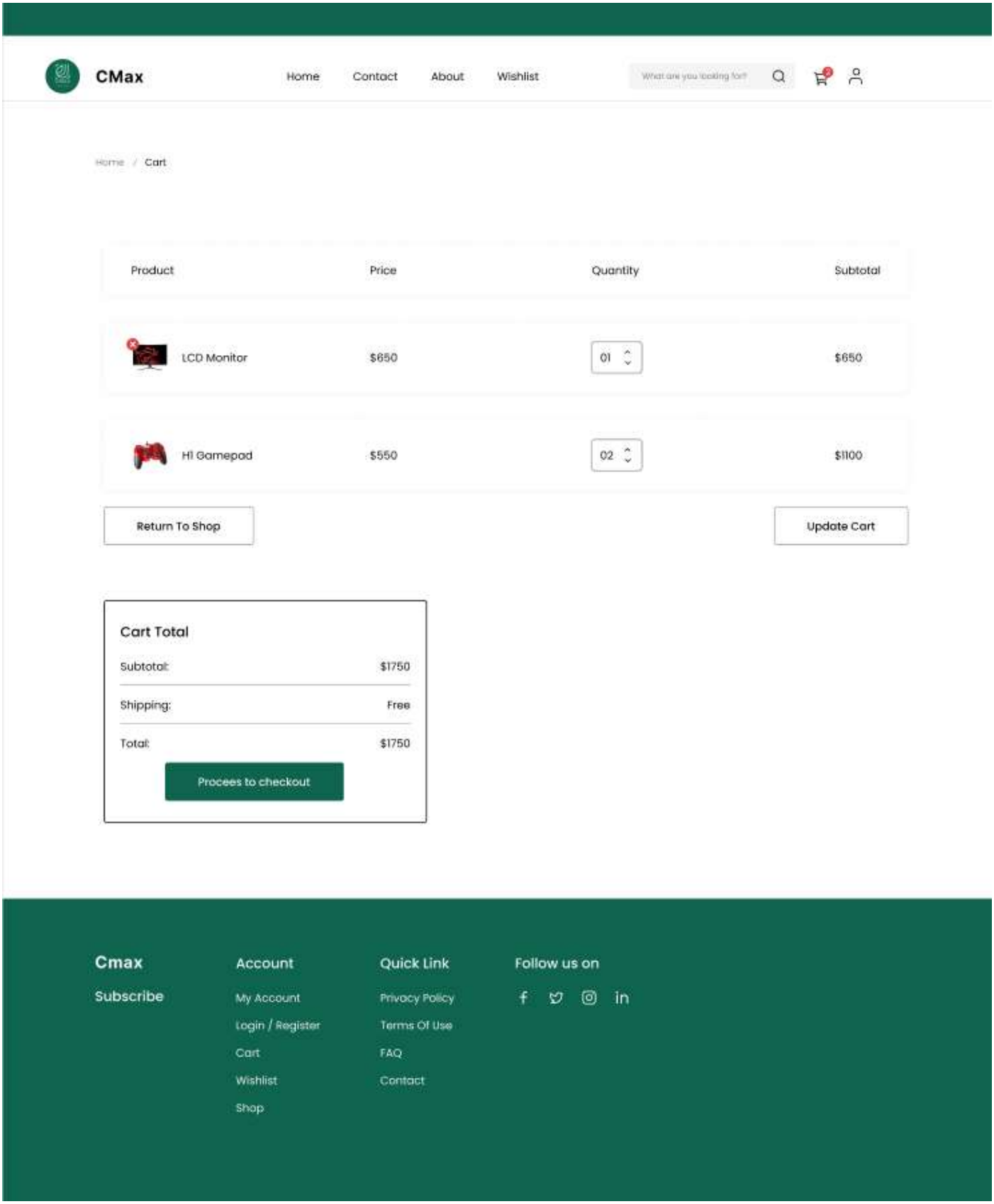

Figure 38 | Cart Page

5.8.1.7 Place Order

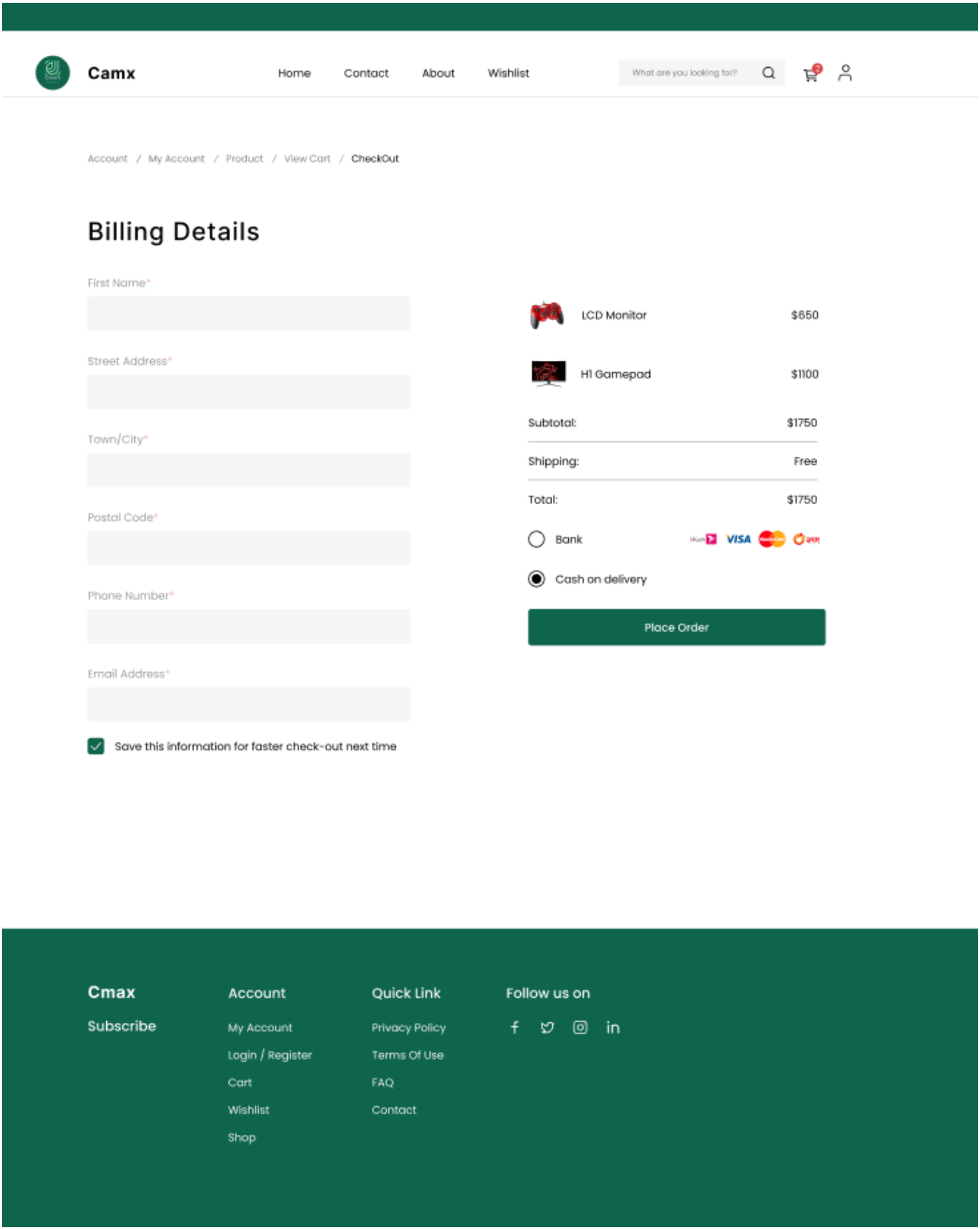

Figure 39 | Place Order Page

5.8.1.8 Account

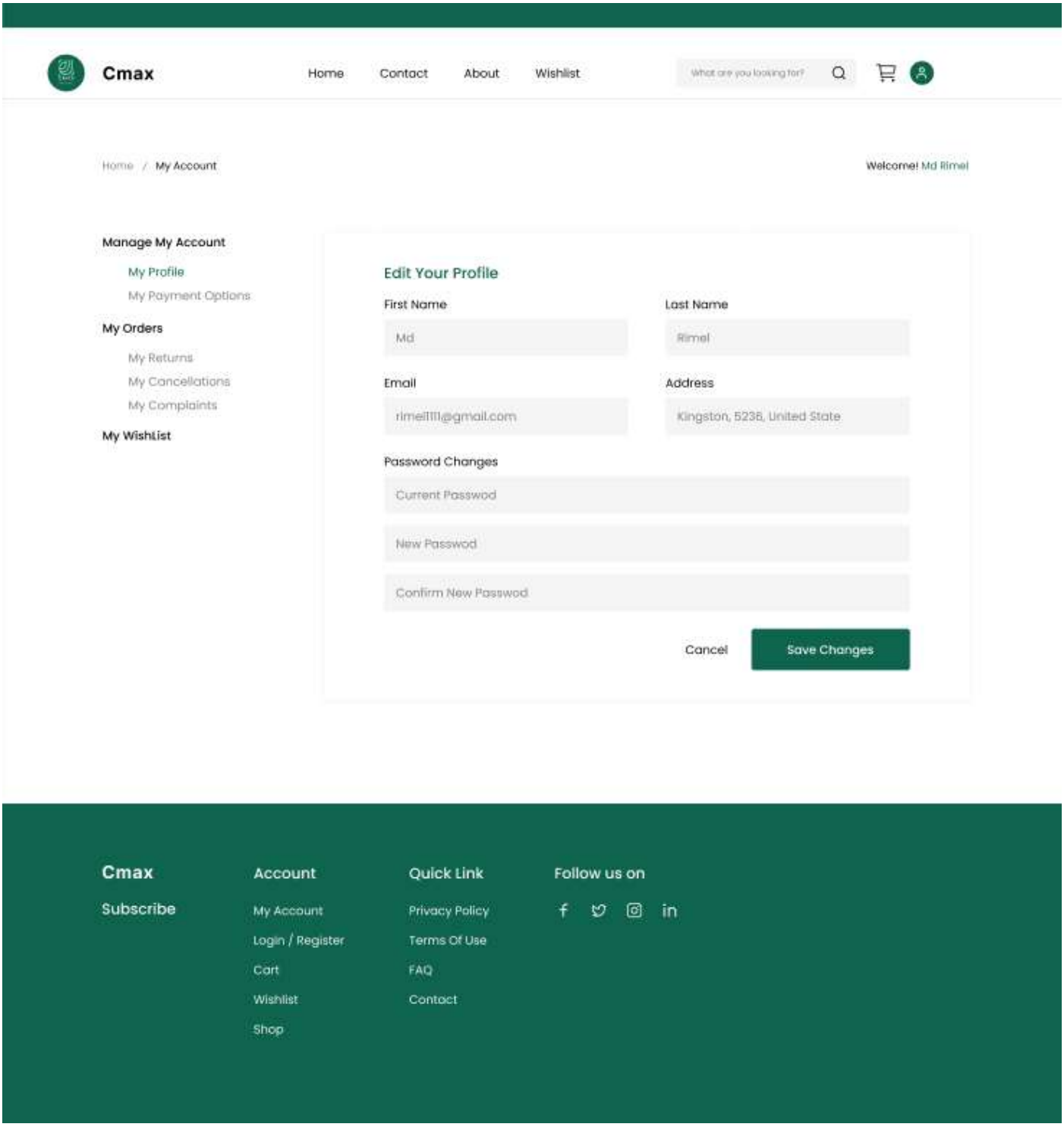

Figure 40 | Account Page

5.8.2 Admin Side

5.8.2.1 Sign Up

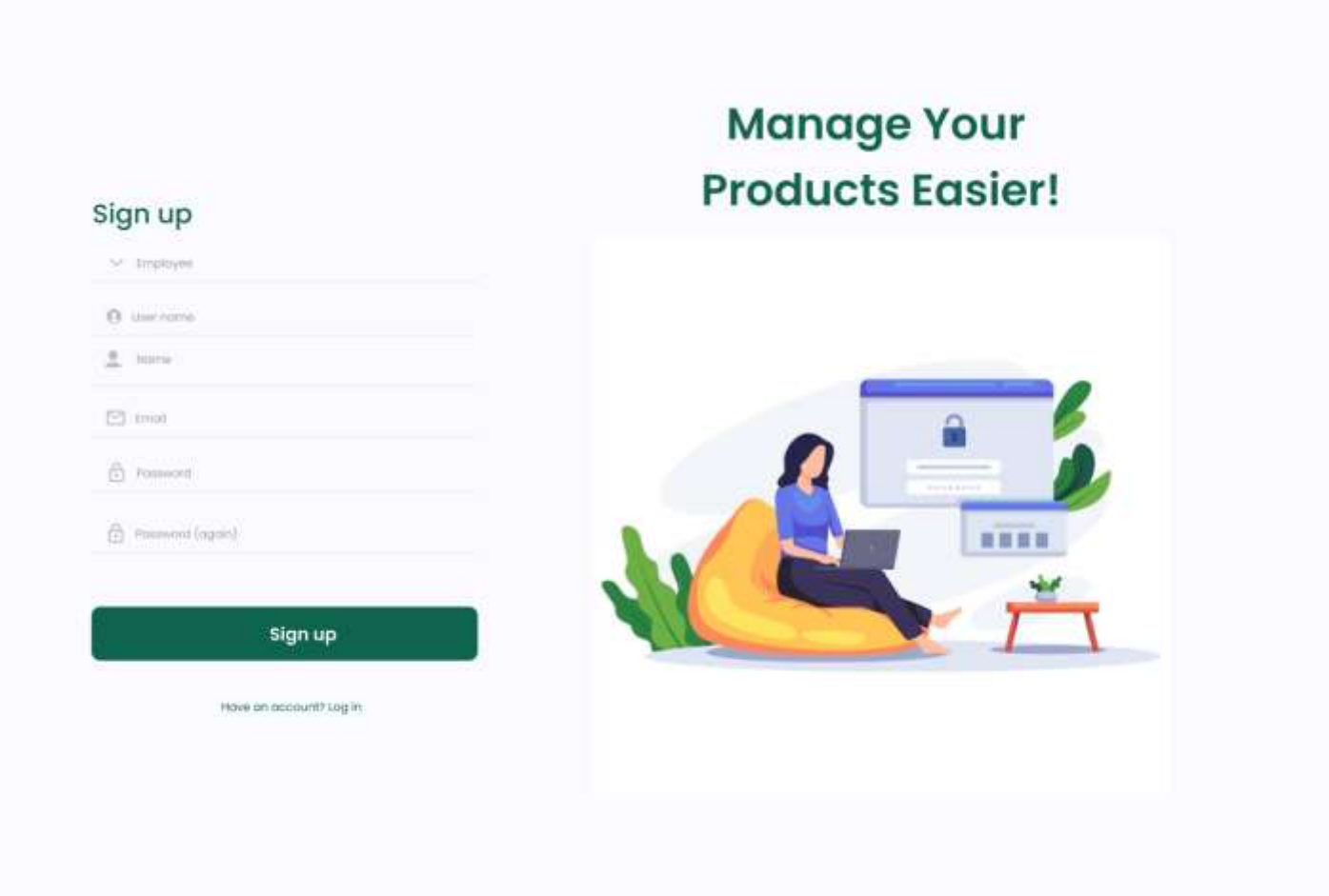

Figure 41 | Sign Up

Figure 42 | Log In

5.8.2.2 Dashboard

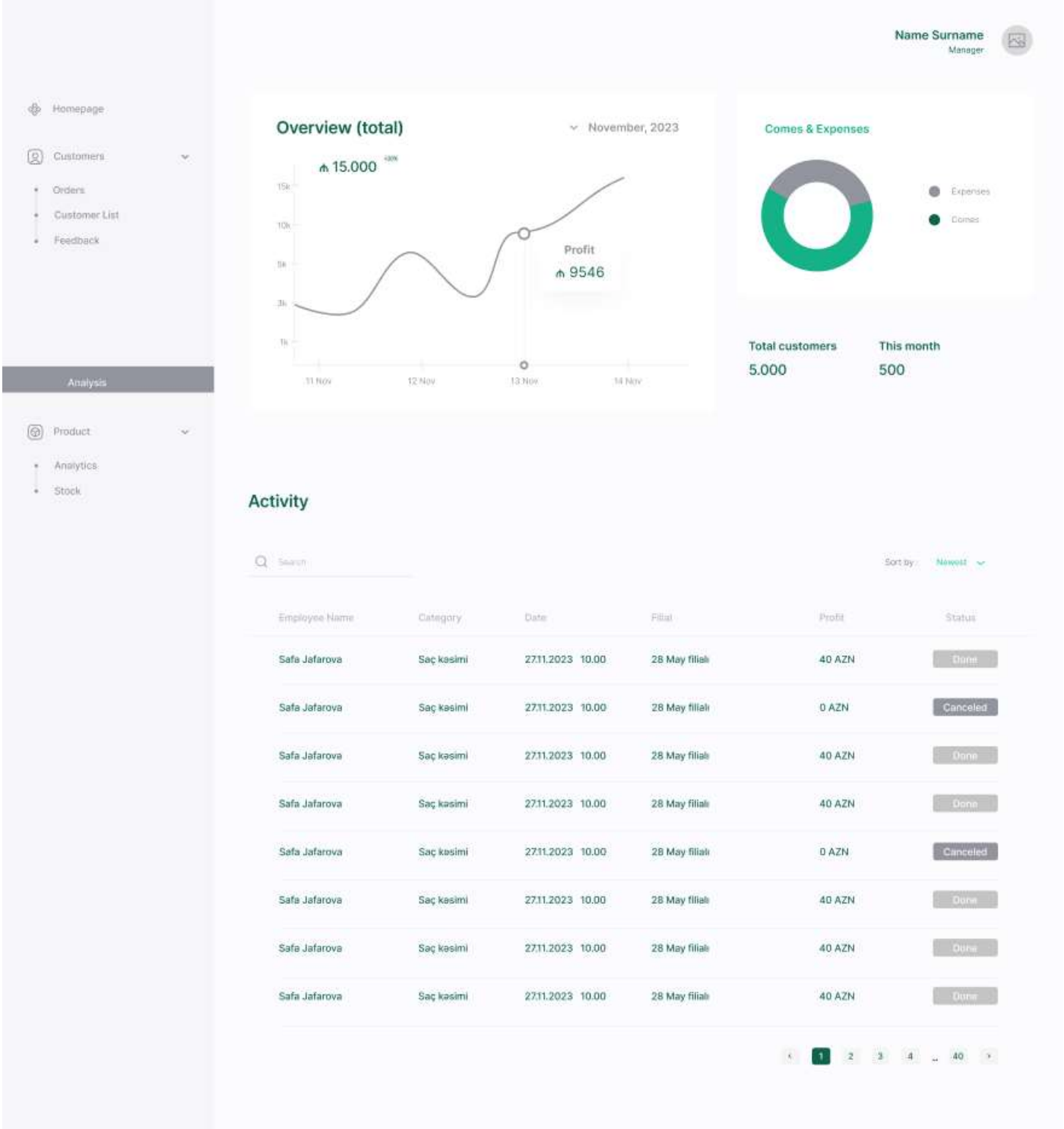

Figure 43 | Dashboard

5.8.2.3 Orders

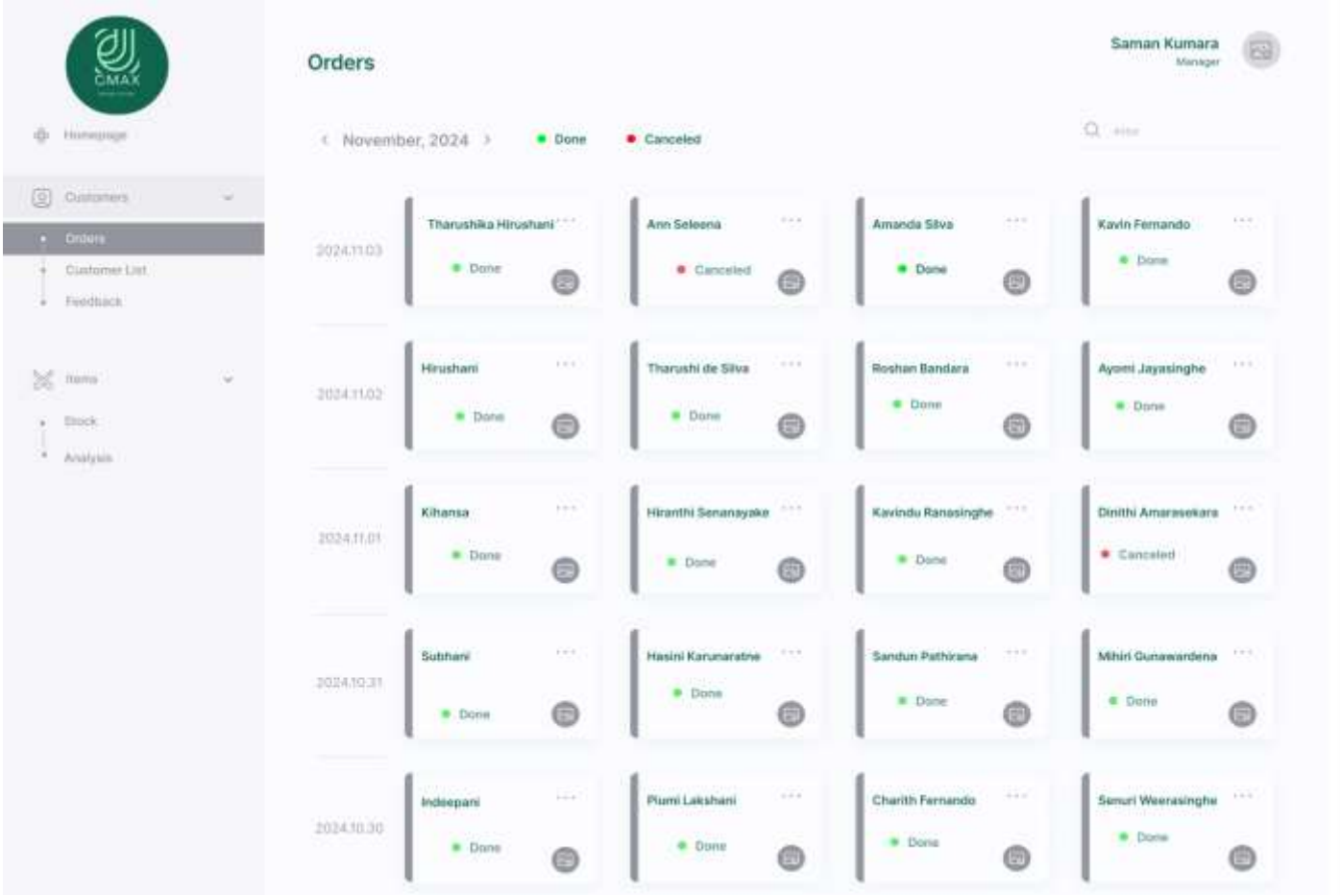

Figure 44 | Orders

5.8.2.4 View Orders

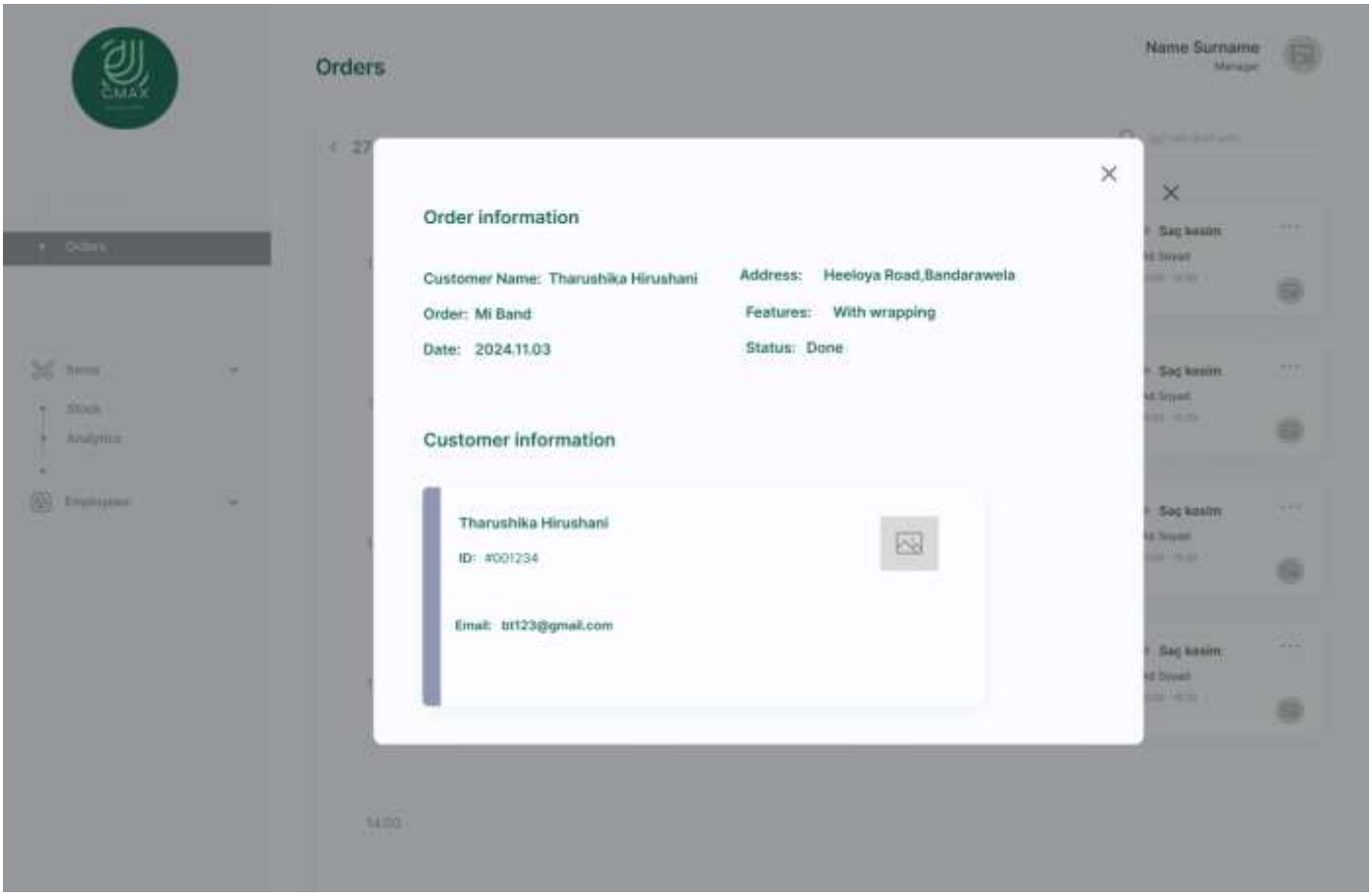

Figure 45 | View Orders

5.8.2.5 Customer List

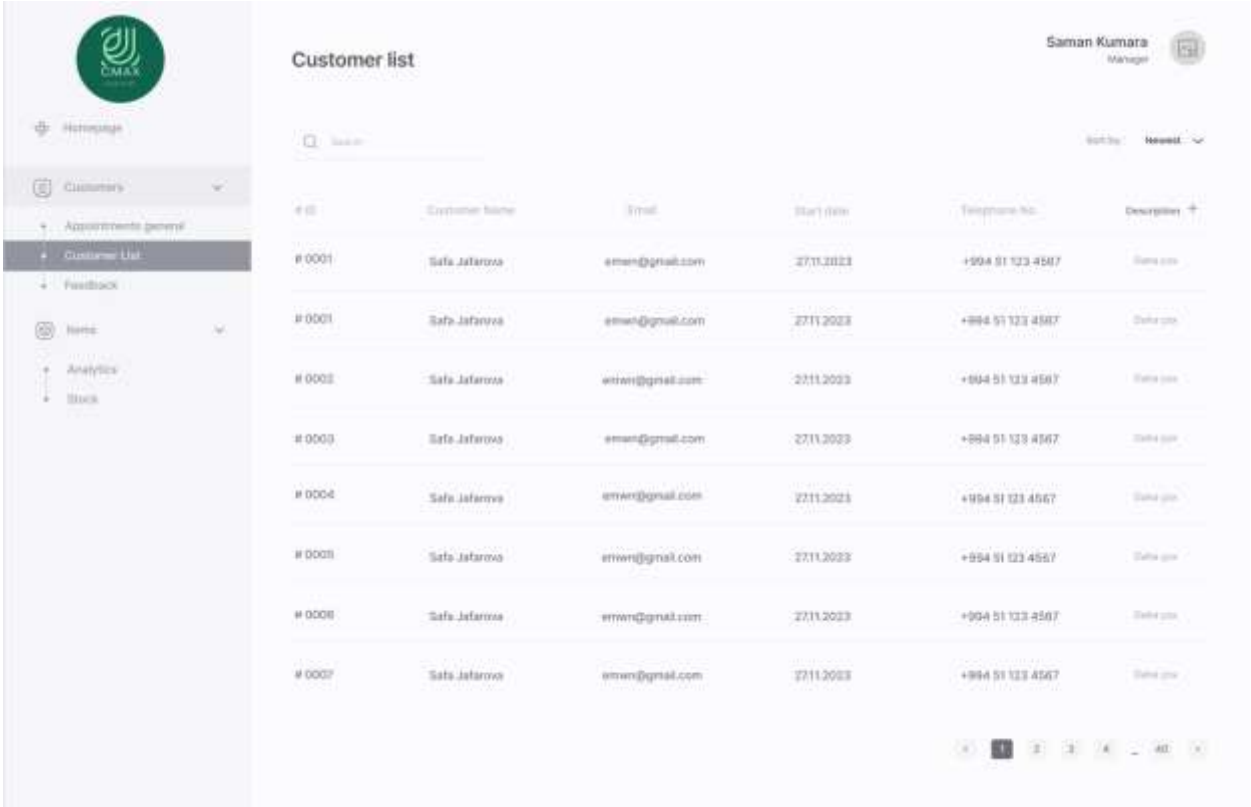

Figure 46 | Customer List

5.8.2.6 Feedback

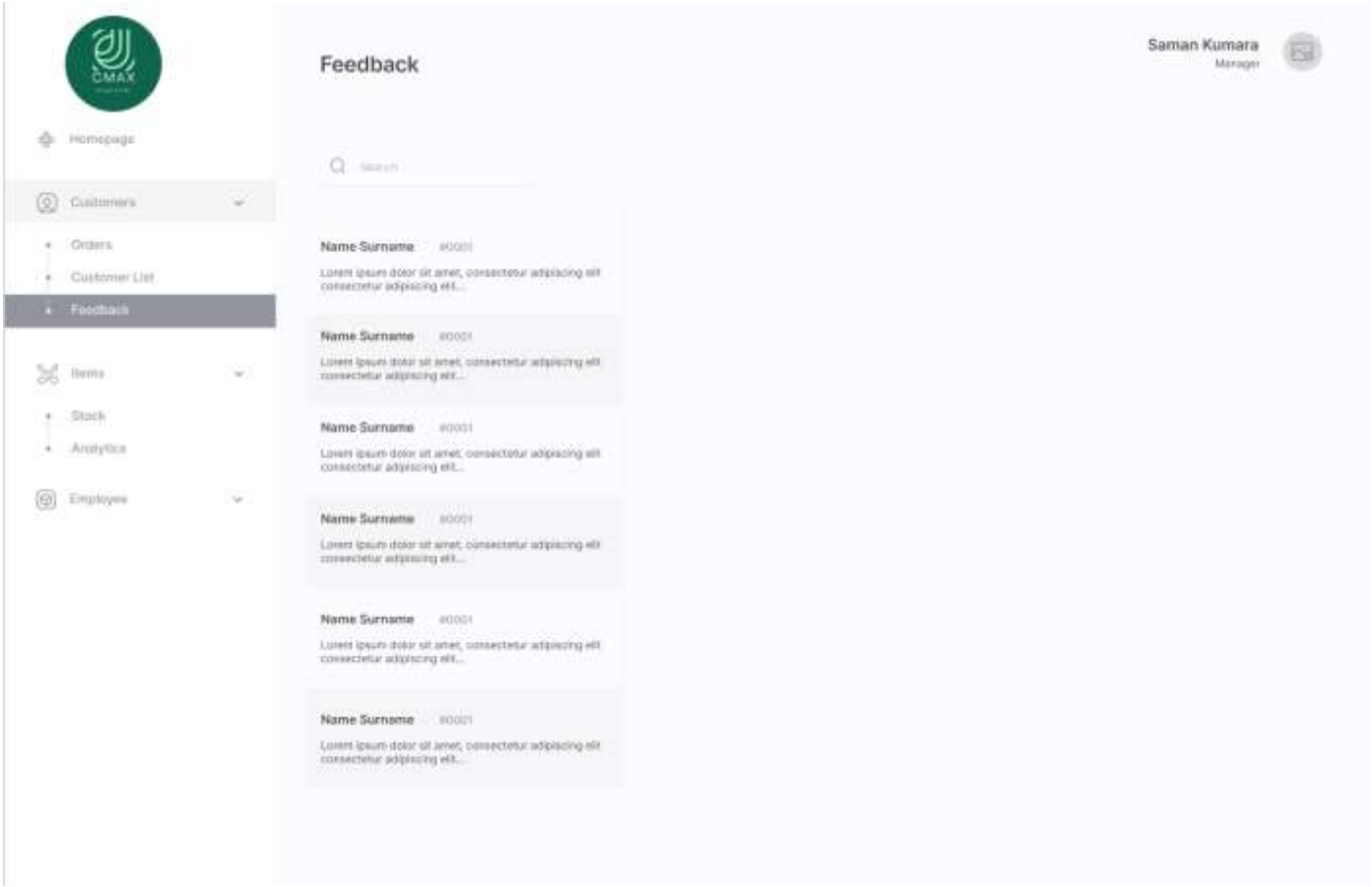

Figure 47 | Feedback

5.8.2.7 Analytics

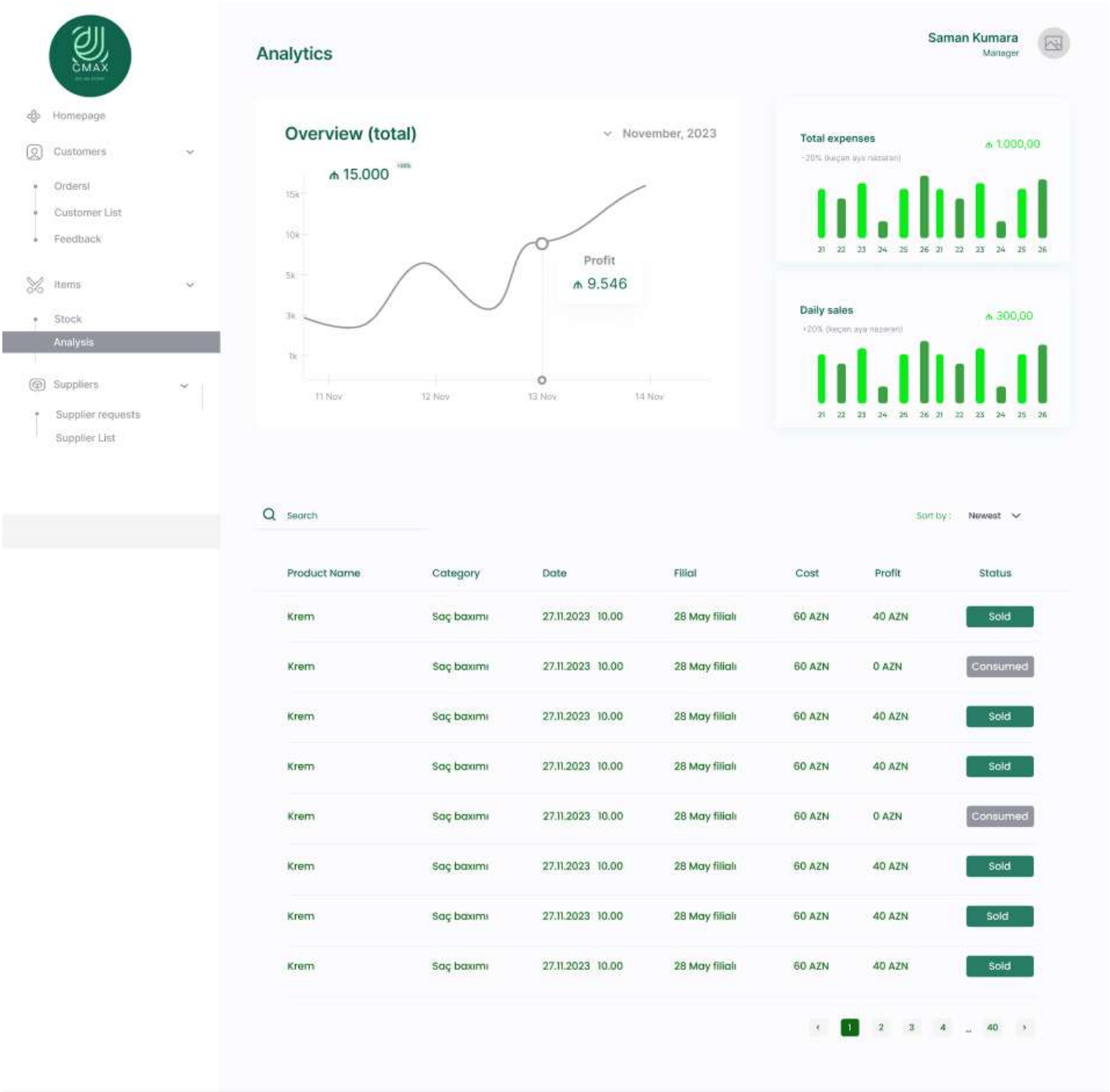

Figure 48 | Analytics

5.8.2.8 Stock

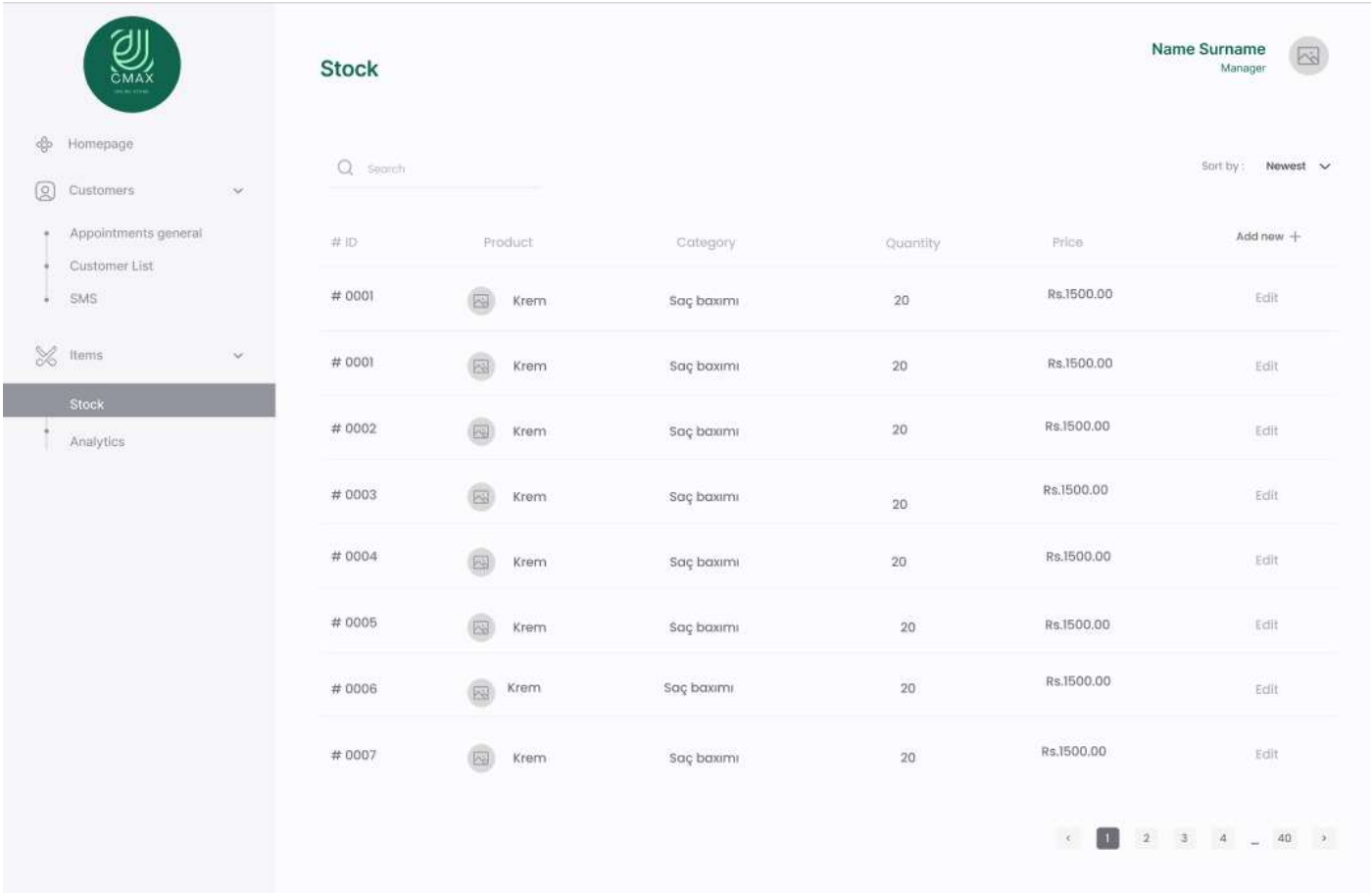

Figure 49 | Stock

5.8.2.9 Add New Product

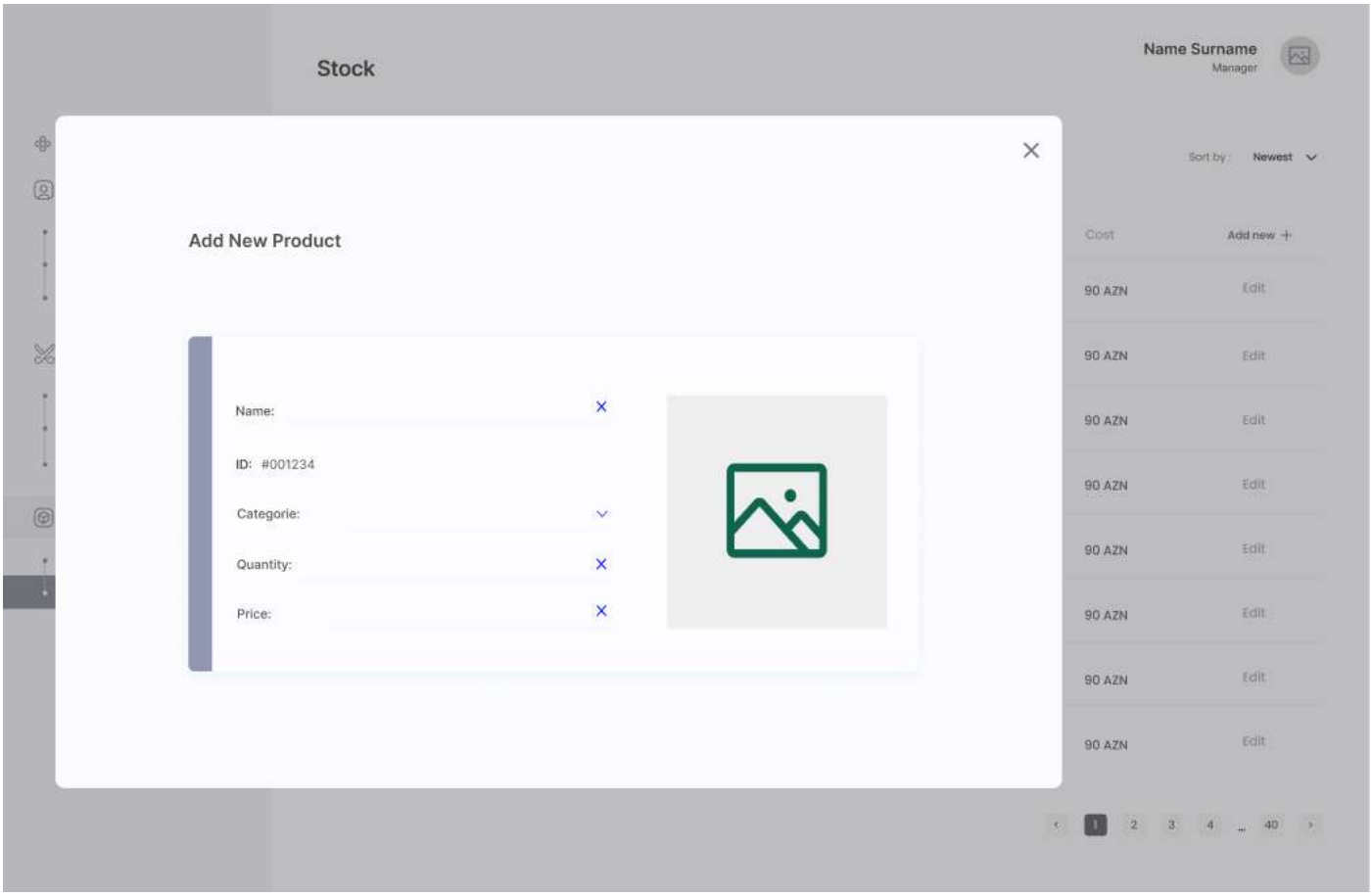

Figure 50 | Add New Product

5.8.2.10 Edit Product Information

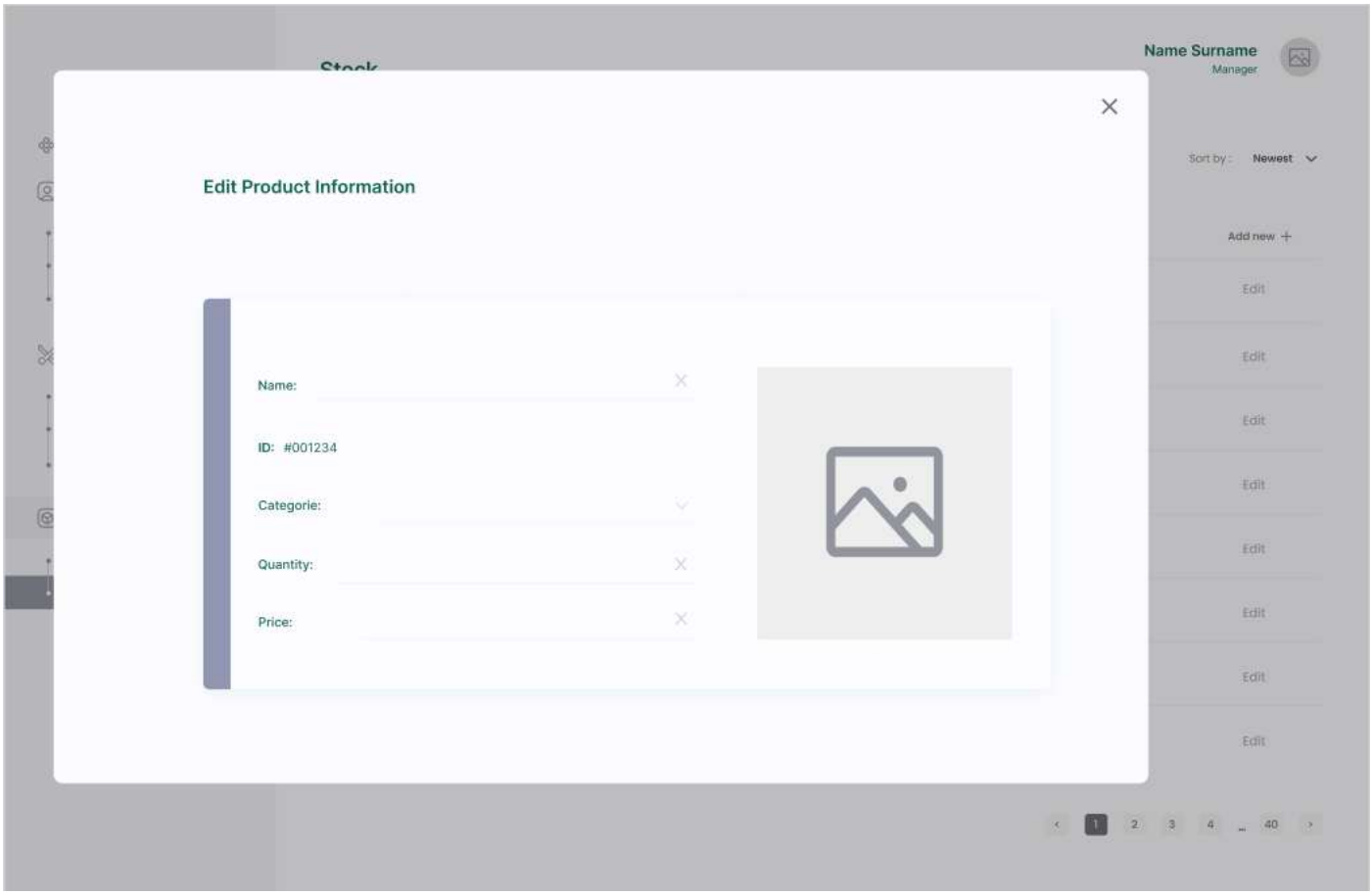

Figure 51 | Edit Product

6. Conclusion

This documentation is for a Web-Based Retail Management System for the CMax Store in Sri Lanka outlines the current system's limitations and the proposed solution to improve operational efficiency and customer experience. The system aims to replace manual processes with a streamlined, automated system that handles customer transactions, inventory, and feedback. Functional requirements include user registration, product browsing, shopping cart, order management, customer feedback, and inventory management. Non-functional requirements focus on system performance, security, scalability, and reliability. The system design includes detailed use case descriptions, user stories, and a Figma-designed user interface.